

Sharper Inner-Gradient Complexity and Parameter-Free Second-Order Methods for Nonconvex–Strongly-Concave Minimax Optimization

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Abstract

We study fully parameter-free second-order methods for smooth nonconvex–strongly-concave minimax optimization. Our analysis uses an existing implicit gradient estimator and its known quadratic accuracy with respect to the inner solution error. Combining this oracle with an inexact universal trust-region framework and a cumulative cubic path analysis, we first obtain a known-parameter method with leading inner-gradient complexity $O((1 + \log \kappa)\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2})$. This removes the multiplicative $\log(1/\epsilon)$ factor in the explicit inner-gradient bound of Minimax Cubic Newton. We then develop a fully parameter-free method based on gradient-scaled trust regions and persistent SCAR tracking. To reach an $O(\epsilon, \sqrt{\kappa^3 \rho} \epsilon)$ second-order stationary point of the primal function $P(x) = \max_y f(x, y)$, its leading trust-region subproblem and inner-gradient complexities are $O(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$ and $O(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2})$, respectively, with additive instance-dependent polylogarithmic terms in $1/\epsilon$. Here Δ_P is the initial primal-function gap, κ the condition number, and ρ the Hessian-Lipschitz constant of f . The parameter-free guarantees hold for sufficiently small ϵ , with an unknown threshold depending only on the regularity constants and uniform over admissible objectives and initializations.

1 Introduction

We consider smooth nonconvex–strongly-concave (NC–SC) minimax optimization

$$\min_{x \in \mathbb{R}^{d_x}} \max_{y \in \mathbb{R}^{d_y}} f(x, y), \quad (1)$$

where $f(x, \cdot)$ is strongly concave for every x , while $f(\cdot, y)$ may be nonconvex. Strong concavity makes the inner maximizer

$$y^*(x) := \arg \max_y f(x, y)$$

well defined and gives rise to the primal function

$$P(x) := \max_y f(x, y).$$

Under standard smoothness assumptions, P is twice differentiable but generally nonconvex. An approximate first-order stationary point of P therefore need not capture local optimality, which motivates the search for approximate second-order stationary points.

Second-order algorithms for NC-SC minimax optimization have been developed using cubic regularization, trust-region methods, and other Newton-type updates (Luo et al., 2022; Chen et al., 2023; Yao and Xu, 2026; Wang and Xu, 2025; Chen et al., 2026). A representative method is Minimax Cubic Newton (MCN) (Luo et al., 2022), which approximates the gradient and Hessian of P through inner maximization and achieves the characteristic $O(\epsilon^{-3/2})$ second-order oracle complexity. Several subsequent methods attain the same $\epsilon^{-3/2}$ -type outer dependence under related second-order stationarity criteria. More recently, joint-minimization reformulations have led to methods that avoid repeatedly solving the inner maximization (Ma et al., 2026a,b).

For methods that construct approximate derivatives of P through inner maximization, however, the outer iteration complexity alone does not describe the total computational work. At each outer iterate x_k , one must approximate $y^*(x_k)$ accurately enough that the resulting first- and second-order information of P can be used reliably. In MCN, for example, accelerated gradient ascent is used for this purpose. Its first-order oracle complexity is stated in $\tilde{O}(\epsilon^{-3/2})$ form, and the explicit bound contains a logarithmic dependence on the target accuracy multiplying the leading outer-order term. This raises a natural question: can one retain the $O(\epsilon^{-3/2})$ second-order iteration dependence while removing the multiplicative logarithmic factor from the total inner-gradient complexity?

We answer this question affirmatively by combining existing primal-function estimators with an analysis of the accumulated inner work. The implicit gradient estimator studied by Ablin et al. (2020) has known quadratic accuracy with respect to the inner solution error; see their Proposition 2.3. The same total-gradient and Schur-complement expressions also appear in Newton-type minimax methods (Zhang et al., 2020). We use these quantities, together with a Newton correction of the value, as oracle ingredients and derive the explicit error bounds needed under our NC-SC assumptions. The complexity improvement comes from combining these bounds with warm starts and a cumulative cubic bound on the outer steps. This path estimate controls the total cost of restoring inner accuracy across successive maximization problems.

Our outer constructions build on the universal trust-region (UTR) framework of Jiang et al. (2026). For the approximate primal-function oracles required here, we also establish a certified-inexact UTR complexity result with additive search overhead; the generic result is deferred to the appendix so that the main text can focus on the minimax problem. In the known-parameter setting, we combine a fixed-scale UTR update with an abstract residual-based inner solver. The inner-work analysis applies to any solver satisfying an accelerated residual-reduction bound, with accelerated gradient descent providing a concrete realization. The resulting method has $O(\epsilon^{-3/2})$ leading total inner-gradient dependence, without a multiplicative $\log(1/\epsilon)$ factor.

We then remove prior knowledge of the problem regularity constants from both the outer and inner components. An adaptive UTR construction combined with the parameter-free SCAR solver yields a fully parameter-free method, but requiring uniformly accurate inner solves reintroduces a multiplicative $\log(1/\epsilon)$ factor in the total inner-gradient complexity. Our final method instead uses a gradient-scaled adaptive trust-region scheme together with persistent SCAR calls across successive inner maximization problems. By retaining the curvature information learned by the inner solver and adapting the inner accuracy to the current outer scale, it recovers the $O(\epsilon^{-3/2})$ leading dependence in both outer and total inner-gradient complexity.

1.1 Main Contributions

Our main contributions are summarized as follows.

Sharper total inner-gradient complexity. Let ℓ and ρ denote the Lipschitz constants of the gradient and Hessian of f , respectively, let μ be the strong concavity constant, and set

$$\kappa = \ell/\mu, \quad \Delta_P = P(x_0) - \inf_x P(x).$$

Algorithm 1 combines the existing implicit gradient and Schur-complement Hessian with a fixed-scale inexact UTR scheme and accesses the inner maximization through an observable gradient-residual

condition. The key estimate is a cumulative cubic bound on the outer steps, which amortizes the logarithmic costs of successive warm-started inner calls. The analysis is stated for a general accelerated residual solver; accelerated gradient descent is one concrete realization. The number of outer trust-region subproblems has leading complexity

$$O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}\right),$$

while the accelerated-gradient realization has leading total inner-gradient complexity

$$O\left((1 + \log \kappa) \Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}\right).$$

All dependence on $\log(1/\epsilon)$ in the full bound is additive rather than multiplicative in the leading $\epsilon^{-3/2}$ term. In particular, this sharpens the explicit first-order oracle bound of MCN with respect to the target accuracy. To our knowledge, this is the first explicit total inner-gradient bound for a second-order NC-SC method that approximates derivatives of the primal function through inner maximization whose leading $O(\epsilon^{-3/2})$ term is not multiplied by a logarithmic factor in $1/\epsilon$. As a technical ingredient, we also establish a certified-inexact UTR complexity result with additive search overhead; its generic statement and proof are deferred to the appendix.

Fully parameter-free second-order methods. We further remove the regularity constants from the executable algorithms. Algorithm 2 combines an adaptive UTR outer scheme with SCAR and requires no knowledge of ℓ , μ , or ρ . It retains the leading $O(\epsilon^{-3/2})$ outer complexity, but uniformly accurate SCAR solves introduce a multiplicative $\log(1/\epsilon)$ factor in the total inner-gradient complexity. Algorithm 3 replaces these independent high-accuracy solves by persistent SCAR tracking within a gradient-scaled adaptive trust-region scheme. For every fixed set of regularity constants and all sufficiently small ϵ , it reaches an $O(\epsilon, \sqrt{\kappa^3 \rho} \epsilon)$ second-order stationary point using

$$O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}\right)$$

trust-region subproblems and

$$O\left(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}\right)$$

inner-gradient evaluations in their leading terms. The remaining dependence on $1/\epsilon$ is additive and polylogarithmic. The executable algorithm uses only the target tolerance and universal numerical constants. The guarantee is class-uniform: for fixed regularity constants, one sufficiently small accuracy threshold applies to every admissible objective and initialization, although this threshold is not known to the algorithm.

Table 1 is intentionally restricted to methods whose stated second-order guarantees and oracle counts are directly comparable to the quantities considered here. Cubic-LocalMinimax (Chen et al., 2023), whose complexity is stated using a different local-minimax stationarity measure, is discussed below but is not reparameterized for the table. Likewise, the recent single-loop regularized Newton method of Ma et al. (2026b) does not use an inner maximization loop of the form measured by the third column and is therefore not included in that comparison.

1.2 Related Work

Implicit differentiation and primal-function estimators. Ablin et al. (2020) analyze the implicit gradient estimator for functions defined through optimization and establish its quadratic error in Proposition 2.3. In minimax optimization, Zhang et al. (2020) use the corresponding total gradient and Schur-complement Hessian; see their equations (1.1)–(1.2). These estimators are ingredients of the present work. Our contributions concern their integration into inexact UTR methods, the accumulated inner-gradient complexity, and parameter-free tracking across changing inner objectives.

Table 1: Complexity comparison of global deterministic second-order methods for smooth NC–SC minimax optimization. Only complexity bounds explicitly stated in the cited papers are reported. Dependence on problem regularity constants is displayed when stated in a directly comparable form; initial optimality gaps and numerical constants are suppressed.

Method	Outer complexity	Total inner-gradient evaluations	Parameter-free
MCN (Luo et al., 2022)	$O(\kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$	$\tilde{O}(\kappa^2 \sqrt{\rho} \epsilon^{-3/2})$	No
MINIMAX-TR / MINIMAX-TRACE (Yao and Xu, 2026)	$O(\epsilon^{-3/2})$	Not reported	No
GRTR / LMNegCur (Wang and Xu, 2025)	$\tilde{O}(\kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$	$\tilde{O}(\kappa^2 \sqrt{\rho} \epsilon^{-3/2})$	No
HSDA / IHSDA (Chen et al., 2026)	$\tilde{O}(\epsilon^{-3/2})$	Not reported	No
Algorithm 1 (ours)	$O(\kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$	$O((1 + \log \kappa) \kappa^2 \sqrt{\rho} \epsilon^{-3/2})$	No
Algorithm 2 (ours)	$O(\kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$	$O(\kappa^2 \sqrt{\rho} \epsilon^{-3/2} \log(1/\epsilon))$	Yes
Algorithm 3 (ours)	$O(\kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$	$O(\kappa^2 \sqrt{\rho} \epsilon^{-3/2})$	Yes

Here $\kappa = \ell/\mu$, and ρ is the Hessian-Lipschitz constant of f . The notation $\tilde{O}(\cdot)$ is retained whenever it is used in the corresponding reference; hidden logarithmic factors in a cited result are not expanded or reconstructed here. “Not reported” means that the method uses inner maximization but the cited work does not state a separate comparable bound on the total number of inner-gradient evaluations. For Algorithms 1–3, the table displays the leading terms of the bounds proved in this paper. The complete bounds for Algorithms 1 and 3 also contain lower-order additive logarithmic or polylogarithmic terms in $1/\epsilon$.

First-order methods for NC–SC minimax optimization. First-order methods for NC–SC minimax problems have been studied extensively. Representative approaches include gradient descent-ascent, multi-step ascent, proximal schemes, and accelerated methods (Nouiehed et al., 2019; Lin et al., 2020a,b). Their complexity for finding first-order stationary points of the primal function has also been studied from both upper- and lower-bound perspectives; see, for example, Zhang et al. (2021) and the references therein. These results form the first-order counterpart to the second-order stationarity problem considered in this paper.

Second-order methods for NC–SC minimax optimization. Luo et al. (2022) proposed MCN, which alternates accelerated gradient ascent for approximating $y^*(x)$ with a cubic-regularized update based on approximate first- and second-order information of the primal function. They established $O(\kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$ second-order oracle complexity and $\tilde{O}(\kappa^2 \sqrt{\rho} \epsilon^{-3/2})$ first-order oracle complexity. Chen et al. (2023) proposed Cubic-LocalMinimax and developed inexact and stochastic variants, using cubic regularization to obtain global convergence toward local minimax points. Their complexity is formulated using a different stationarity measure, so we do not convert it to the accuracy parameterization used in Table 1.

Trust-region approaches were developed by Yao and Xu (2026), who proposed MINIMAX-TR and MINIMAX-TRACE. Wang and Xu (2025) subsequently proposed gradient-norm regularized trust-region and Levenberg–Marquardt methods, together with inexact variants, and established $\tilde{O}(\kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2})$ outer complexity and $\tilde{O}(\kappa^2 \sqrt{\rho} \epsilon^{-3/2})$ total gradient-ascent complexity. More recently, Chen et al. (2026) proposed homogeneous second-order descent-ascent methods and developed an inexact Lanczos implementation for large-scale problems.

A recent alternative is to avoid repeated inner maximization altogether. Ma et al. (2026a) introduced an equivalent joint-minimization reformulation that preserves first- and second-order stationarity and used it to develop a parameter-free line-search framework. Building on this reformulation, Ma et al. (2026b) proposed a single-loop regularized Newton method with $O(\epsilon^{-3/2})$ global iteration complexity and local superlinear convergence. This line of work addresses the cost of the inner loop by changing the formulation, whereas the present paper studies the total inner-gradient complexity when approximate derivatives of the primal function are constructed through inner maximization.

Adaptive and parameter-free optimization. Adaptive optimization methods aim to reduce or eliminate prior knowledge of problem regularity constants. In smooth nonconvex optimization, the universal trust-region (UTR) framework (Jiang et al., 2026) uses gradient-scaled trust-region updates together with adaptive scale adjustment, and provides the outer optimization framework on which our methods are built. On the inner side, Lan et al. (2026) developed SCAR, a parameter-free accelerated method for smooth strongly convex optimization that adapts to unknown smoothness and strong-convexity parameters while retaining accelerated gradient complexity. These two ingredients play different roles in our development: UTR provides the outer second-order mechanism, whereas SCAR provides a parameter-free realization of the strongly concave inner maximization.

Related adaptive and parameter-free methods have also been developed for minimax optimization and monotone variational inequalities (Jiang et al., 2024; Jiang and Mokhtari, 2025). More recently, Wang et al. (2026) proposed a fully parameter-free second-order method for convex–concave minimax optimization, requiring neither the Hessian-Lipschitz constant nor an a priori distance bound to a solution. For NC–SC minimax problems, Ma et al. (2026a) developed a parameter-free line-search framework based on a joint-minimization reformulation. In contrast, our constructions retain the primal-function formulation: Algorithm 2 combines adaptive UTR with SCAR, while Algorithm 3 couples a gradient-scaled adaptive trust-region scheme with persistent SCAR tracking across successive inner maximization problems.

1.3 Organization

The remainder of the paper is organized as follows. Section 2 presents the problem setting, properties of the primal function, the inner residual interface, and the Newton-corrected estimators. Section 3 develops Algorithm 1, a known-parameter UTR method with a general accelerated inner residual solver, and establishes its outer and total inner-gradient complexity. Section 4 first combines adaptive UTR with SCAR in Algorithm 2 and then develops the persistent-SCAR gradient-scaled trust-region method, Algorithm 3. Section ?? presents the numerical experiments, and Section ?? concludes the paper. The generic certified-inexact UTR results and deferred technical proofs are collected in the appendices.

2 Preliminaries

This section collects the notation and analytical ingredients used throughout the paper. We first recall standard properties of the primal function associated with smooth NC–SC minimax optimization. We then formulate the inner maximization through an observable gradient-residual interface and record quantitative bounds for the primal-function estimators used below. Finally, we record a generic inexact trust-region estimate that will be repeatedly specialized in the subsequent sections.

2.1 Problem Setting and Standard Properties

We consider the smooth NC–SC minimax problem

$$\min_{x \in \mathbb{R}^{d_x}} P(x), \quad P(x) := \max_{y \in \mathbb{R}^{d_y}} f(x, y). \quad (2)$$

For every x , define

$$y^*(x) := \arg \max_{y \in \mathbb{R}^{d_y}} f(x, y). \quad (3)$$

We write

$$P^* := \inf_x P(x), \quad \Delta_P := P(x_0) - P^*.$$

Throughout the paper, $\|\cdot\|$ denotes the Euclidean norm for vectors and the spectral norm for matrices. For a symmetric matrix A , $\lambda_{\min}(A)$ denotes its smallest eigenvalue.

Assumption 1 (NC–SC regularity) *The payoff $f : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}$ is twice continuously differentiable and satisfies:*

- (i) ∇f is ℓ -Lipschitz;
- (ii) $\nabla^2 f$ is ρ -Lipschitz;
- (iii) $f(x, \cdot)$ is μ -strongly concave for every x , where $0 < \mu \leq \ell$;
- (iv) $P^* > -\infty$.

We write

$$\kappa := \frac{\ell}{\mu}. \quad (4)$$

Under Assumption 1, the maximizer $y^*(x)$ is unique and $\nabla_{yy}^2 f(x, y)$ is nonsingular. Define the Schur-complement map

$$\mathcal{H}(x, y) := \nabla_{xx}^2 f(x, y) - \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} \nabla_{yx}^2 f(x, y). \quad (5)$$

We use the following standard properties of the primal function.

Lemma 2 (Primal-function geometry) *Under Assumption 1, $y^*(\cdot)$ is κ -Lipschitz and P is twice continuously differentiable with*

$$\nabla P(x) = \nabla_x f(x, y^*(x)), \quad (6a)$$

$$\nabla^2 P(x) = \mathcal{H}(x, y^*(x)). \quad (6b)$$

Moreover, ∇P is $(1 + \kappa)\ell$ -Lipschitz and

$$\|\nabla^2 P(x) - \nabla^2 P(x')\| \leq M_P \|x - x'\|, \quad M_P := 4\sqrt{2} \kappa^3 \rho. \quad (7)$$

These properties follow from Luo et al. (2022, Lemmas 1–3).

Definition 3 (Approximate second-order stationarity) *For $\epsilon_g, \epsilon_H > 0$, a point x is an (ϵ_g, ϵ_H) -second-order stationary point of P if*

$$\|\nabla P(x)\| \leq \epsilon_g, \quad \lambda_{\min}(\nabla^2 P(x)) \geq -\epsilon_H. \quad (8)$$

Our main target is an $(\epsilon, \sqrt{M_P \epsilon})$ -second-order stationary point.

2.2 Inner Residuals and Primal-Function Estimators

The distance $\|y - y^*(x)\|$ is useful analytically but is not directly observable. We therefore formulate the inner maximization in terms of the gradient residual $\|\nabla_y f(x, y)\|$.

Lemma 4 (Residual–distance certificate) *Under Assumption 1, for every x and y ,*

$$\|y - y^*(x)\| \leq \frac{\|\nabla_y f(x, y)\|}{\mu} \leq \kappa \|y - y^*(x)\|. \quad (9)$$

Proof The first inequality follows from strong concavity and the Cauchy–Schwarz inequality. The second follows from ℓ -smoothness and $\nabla_y f(x, y^*(x)) = 0$. ■

We use the inner maximization only through the following interface.

Definition 5 (Inner residual module) *Given an outer point x , a warm start y^{in} , and a target $\tau > 0$, an inner residual module $\mathcal{S}$ returns*

$$y^{\text{out}} = \text{MaxRes}_{\mathcal{S}}(x, y^{\text{in}}, \tau) \quad \text{such that} \quad \|\nabla_y f(x, y^{\text{out}})\| \leq \tau. \quad (10)$$

We call $\mathcal{S}$ accelerated if there exist $C_{\mathcal{S}} > 0$ and $b_{\mathcal{S}} \geq 1$, which may depend on the implementation and on κ but not on x , τ , the initial residual, ϵ , or the call index, such that a call with

$$R_{\text{in}} := \|\nabla_y f(x, y^{\text{in}})\|$$

uses at most

$$C_{\mathcal{S}} \sqrt{\kappa} \left[b_{\mathcal{S}} + \left\lceil \log_2 \max \left\{ 1, \frac{R_{\text{in}}}{\tau} \right\} \right\rceil \right] \quad (11)$$

evaluations of $\nabla_y f$.

The residual condition (10) is the only inner-solver property needed for outer correctness. The work bound (11) is used only when deriving total inner-gradient complexity. Concrete realizations by accelerated gradient descent and SCAR are introduced in Sections 3 and 4, respectively.

We count one evaluation of $\nabla_y f(x, y)$ as one inner-gradient evaluation. Corrected-gradient and Schur-complement constructions, trust-region subproblem solves, and linear algebra used to apply $(\nabla_{yy}^2 f)^{-1}$ are accounted for separately.

We use the following value, gradient, and Hessian approximations at an approximate inner maximizer y . Define

$$\hat{P}(x, y) := f(x, y) - \frac{1}{2} \nabla_y f(x, y)^\top (\nabla_{yy}^2 f(x, y))^{-1} \nabla_y f(x, y), \quad (12)$$

and

$$\hat{g}(x, y) := \nabla_x f(x, y) - \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} \nabla_y f(x, y), \quad (13a)$$

$$\hat{H}(x, y) := \mathcal{H}(x, y). \quad (13b)$$

Since $\nabla_{yy}^2 f(x, y) \prec 0$, the correction in (12) raises the raw payoff and is exact for a concave quadratic inner problem.

The gradient in (13a) is the existing implicit estimator studied by Ablin et al. (2020), adapted to inner maximization. Its quadratic accuracy is established in their Proposition 2.3. The same total-gradient expression and the Schur-complement Hessian appear in Zhang et al. (2020). We use these oracle ingredients and the value correction to obtain the explicit bounds required by the complexity analysis. MCN instead uses the raw plug-in gradient $\nabla_x f(x, y)$, while both methods use the Schur-complement Hessian.

Lemma 6 (Quantitative primal-function oracle bounds) *Under Assumption 1, for every x and y ,*

$$|\hat{P}(x, y) - P(x)| \leq \frac{(3\kappa - 1)\rho}{12} \|y - y^*(x)\|^3 \leq \frac{\kappa\rho}{4} \|y - y^*(x)\|^3, \quad (14a)$$

$$\|\hat{g}(x, y) - \nabla P(x)\| \leq \frac{(1 + \kappa)\rho}{2} \|y - y^*(x)\|^2, \quad (14b)$$

$$\|\hat{H}(x, y) - \nabla^2 P(x)\| \leq (1 + \kappa)^2 \rho \|y - y^*(x)\|. \quad (14c)$$

Moreover, whenever

$$\|y - y^*(x)\| \leq \frac{\mu}{\rho}, \quad (15)$$

the corrected value satisfies

$$|\hat{P}(x, y) - P(x)| \leq \frac{7\rho}{24} \|y - y^*(x)\|^3. \quad (16)$$

The proof is given in Appendix A.1. For completeness, we recall the cancellation behind the known quadratic gradient error in (14b). With

$$y_t := y^*(x) + t(y - y^*(x)), \quad t \in [0, 1],$$

the proof derives the exact identity

$$\begin{aligned} \hat{g}(x, y) - \nabla P(x) &= \int_0^1 \left[\nabla_{xy}^2 f(x, y_t) - \nabla_{xy}^2 f(x, y) \right. \\ &\quad \left. + \nabla_{xy}^2 f(x, y) (\nabla_{yy}^2 f(x, y))^{-1} (\nabla_{yy}^2 f(x, y) - \nabla_{yy}^2 f(x, y_t)) \right] \\ &\quad \times (y - y^*(x)) dt. \end{aligned} \quad (17)$$

Hessian Lipschitz continuity then gives the quadratic bound in (14b).

For comparison, the raw plug-in gradient satisfies

$$\|\nabla_x f(x, y) - \nabla P(x)\| \leq \ell \|y - y^*(x)\|. \quad (18)$$

Thus an $O(\epsilon)$ gradient approximation based on the raw estimator requires an inner distance of order $O(\epsilon)$, whereas the corrected gradient only requires order $O(\sqrt{\epsilon})$.

Combining Lemmas 4 and 6 gives an observable form of the same estimates.

Corollary 7 (Residual-based estimator bounds) *Let*

$$R_y(x, y) := \|\nabla_y f(x, y)\|.$$

Under Assumption 1,

$$|\hat{P}(x, y) - P(x)| \leq \frac{(3\kappa - 1)\rho}{12\mu^3} R_y(x, y)^3, \quad (19a)$$

$$\|\hat{g}(x, y) - \nabla P(x)\| \leq \frac{(1 + \kappa)\rho}{2\mu^2} R_y(x, y)^2, \quad (19b)$$

$$\|\hat{H}(x, y) - \nabla^2 P(x)\| \leq \frac{(1 + \kappa)^2 \rho}{\mu} R_y(x, y). \quad (19c)$$

If

$$R_y(x, y) \leq \frac{\mu^2}{\rho}, \quad (20)$$

then

$$|\hat{P}(x, y) - P(x)| \leq \frac{7\rho}{24\mu^3} R_y(x, y)^3. \quad (21)$$

Proof By Lemma 4, $\|y - y^*(x)\| \leq R_y(x, y)/\mu$. Substitution into Lemma 6 proves the result. $\blacksquare$

2.3 Trust-Region Preliminaries

We finally record the generic trust-region estimates used by the outer methods. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable and suppose that its Hessian is M -Lipschitz:

$$\|\nabla^2 F(x) - \nabla^2 F(z)\| \leq M\|x - z\|. \quad (22)$$

The following Taylor estimates are standard.

Lemma 8 (Second-order Taylor bounds) *For all $x, d \in \mathbb{R}^n$,*

$$\|\nabla F(x + d) - \nabla F(x) - \nabla^2 F(x)d\| \leq \frac{M}{2}\|d\|^2, \quad (23a)$$

$$F(x + d) \leq F(x) + \nabla F(x)^\top d + \frac{1}{2}d^\top \nabla^2 F(x)d + \frac{M}{6}\|d\|^3. \quad (23b)$$

These estimates are standard; see, e.g., Nesterov (2004, Lemma 1.2.4).

We also use the classical characterization of a globally solved quadratic trust-region subproblem.

Lemma 9 (Global optimality for a quadratic trust-region problem) *Let $g \in \mathbb{R}^n$, let $B \in \mathbb{R}^{n \times n}$ be symmetric, and let $R > 0$. A vector d is a global solution of*

$$\min_{\|u\| \leq R} g^\top u + \frac{1}{2}u^\top Bu \quad (24)$$

if and only if there exists $\lambda \geq 0$ such that

$$(B + \lambda I)d = -g, \quad B + \lambda I \succeq 0, \quad \|d\| \leq R, \quad \lambda(\|d\| - R) = 0. \quad (25)$$

In particular, $\lambda > 0$ implies $\|d\| = R$.

This characterization is classical; see, e.g., Luenberger and Ye (2021, Section 8.7).

The next estimate is the common one-step inequality underlying the fixed-scale and adaptive UTR analyses as well as the parameter-free trust-region constructions later in the paper.

Lemma 10 (Common inexact trust-region step estimate) *Suppose (22) holds. At a point x , let g and a symmetric matrix H satisfy*

$$\|g - \nabla F(x)\| \leq \delta_0, \quad \|H - \nabla^2 F(x)\| \leq \xi. \quad (26)$$

For $a \geq 0$ and $R > 0$, let d be a global solution of

$$\min_{\|u\| \leq R} q(u) := g^\top u + \frac{1}{2}u^\top (H + aI)u, \quad (27)$$

let λ be its trust-region multiplier, and write $r := \|d\|$. Let a candidate gradient g^+ satisfy

$$\|g^+ - \nabla F(x + d)\| \leq \delta_+. \quad (28)$$

Then

$$q(d) \leq -\frac{\lambda}{2}r^2, \quad (29a)$$

$$F(x) - F(x + d) \geq \frac{a - \xi}{2}r^2 + \frac{\lambda}{2}r^2 - \frac{M}{6}r^3 - \delta_0 r, \quad (29b)$$

$$\|g^+\| \leq (a + \xi + \lambda)r + \frac{M}{2}r^2 + \delta_0 + \delta_+. \quad (29c)$$

The proof is deferred to Appendix A.2.

3 A Trust-Region Method with Sharper Inner-Gradient Complexity

We first consider the setting in which the regularity constants ℓ , μ , and ρ are available to the outer method. The analysis has two ingredients. We begin with a fixed-scale inexact estimate for additive universal trust-region (UTR) steps. We then apply this estimate to the primal function using the Newton-corrected estimators from Section 2. The resulting outer complexity is standard in its $\epsilon^{-3/2}$ dependence; the main point of this section is that the total work of the inner maximization can be accumulated without multiplying this leading term by $\log(1/\epsilon)$.

We use

$$\log_+(t) := \max\{\log t, 0\}, \quad t > 0,$$

and set $\log_+(0) := 0$.

3.1 A Fixed-Scale Inexact UTR Estimate

We first record the generic outer estimate used below. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable, bounded below by

$$F^* := \inf_x F(x) > -\infty,$$

and suppose that its Hessian is M -Lipschitz, as in (22).

The following procedure is a certified-inexact extension of the fixed-scale additive UTR update of Jiang et al. (2026). We use it only as an outer analytical interface for the minimax method below.

Assumption 11 (Fixed-scale inexact oracle) *Fix $\epsilon > 0$. At every iterate x_k , the oracle returns $g_k \in \mathbb{R}^n$, a symmetric matrix $H_k \in \mathbb{R}^{n \times n}$, and an observable certificate $\xi_k \geq 0$ satisfying*

$$\|g_k - \nabla F(x_k)\| \leq \frac{\epsilon}{8192}, \quad \|H_k - \nabla^2 F(x_k)\| \leq \xi_k \leq \frac{1}{64} \sqrt{\frac{M}{6}} \sqrt{\epsilon}. \quad (30)$$

Set

$$h_k := \max\{\|g_k\|, \epsilon\}. \quad (31)$$

At x_k , terminate if

$$\|g_k\| \leq \epsilon, \quad \lambda_{\min}(H_k) \geq -\sqrt{M}\epsilon + \xi_k. \quad (32)$$

Otherwise, compute a global solution d_k of

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \quad & g_k^\top d + \frac{1}{2} d^\top \left(H_k + \xi_k I + 2\sqrt{\frac{M}{6}} \sqrt{h_k} I \right) d, \\ \text{s. t.} \quad & \|d\| \leq \frac{\sqrt{h_k}}{4\sqrt{M/6}}, \end{aligned} \quad (33)$$

and set $x_{k+1} = x_k + d_k$.

The shift $\xi_k I$ compensates for the certified Hessian error in both the model and the stopping test. The subproblem (33) retains the gradient-scaled regularization and radius of additive UTR, while allowing inexact first- and second-order information.

Theorem 12 (Fixed-scale inexact UTR estimate) *Fix x_0 and $\epsilon > 0$, define*

$$\Delta_F := F(x_0) - F^*, \quad h_0 := \max\{\|g_0\|, \epsilon\},$$

and suppose Assumption 11 holds. Repeated application of (32)–(33) terminates after at most

$$K \leq 65 \sqrt{\frac{M}{6}} \Delta_F \epsilon^{-3/2} + 3 \log_+ \frac{h_0}{\epsilon} + 2 \quad (34)$$

nonterminal iterations, equivalently trust-region subproblem solves. In particular,

$$K = O\left(\Delta_F \sqrt{M} \epsilon^{-3/2} + \log_+ \frac{h_0}{\epsilon} + 1\right).$$

The returned point $\hat{x}$ satisfies

$$\|\nabla F(\hat{x})\| \leq \frac{8193}{8192} \epsilon, \quad \lambda_{\min}(\nabla^2 F(\hat{x})) \geq -\sqrt{M} \epsilon. \quad (35)$$

Consequently, replacing the internal gradient tolerance by $8192\epsilon/8193$ gives the standard $(\epsilon, \sqrt{M}\epsilon)$ guarantee without changing the complexity order.

The proof is given in Appendix B.1. Unlike the exact-oracle case, an inexact interior step need not decrease F monotonically. The proof controls this possible increase through a finite-prefix argument; we defer this technical accounting to the appendix.

For the inner-work analysis below, we need one additional consequence of the same fixed-scale argument.

Lemma 13 (Cubic path budget) *Under the assumptions of Theorem 12, let $r_k := \|d_k\|$, and let K denote the return index. Then*

$$M \sum_{k=0}^{K-1} r_k^3 \leq 4\Delta_F + \frac{K}{98\,304\sqrt{6}} \frac{\epsilon^{3/2}}{\sqrt{M}}. \quad (36)$$

Theorem 12 controls the number of outer subproblems. Lemma 13 provides a different type of information: it controls the cumulative motion of the outer iterates. This path estimate is what allows the logarithmic costs of the inner maximization to be amortized over the full trajectory rather than bounded independently at every iteration.

3.2 A Known-Parameter NC–SC Trust-Region Method

We now specialize the fixed-scale estimate to

$$P(x) = \max_y f(x, y).$$

Recall from Lemma 2 that

$$M_P = 4\sqrt{2} \kappa^3 \rho$$

is a valid Hessian-Lipschitz constant for P . We choose the inner residual tolerance so that the corrected estimators satisfy Assumption 11 with $F = P$ and $M = M_P$.

Set

$$\tau_{\text{out}} := \frac{\mu \sqrt{M_P \epsilon / 6}}{64(1 + \kappa)^2 \rho}. \quad (37)$$

This target is expressed in terms of the observable residual $\|\nabla_y f(x, y)\|$. In contrast, the MCN analysis of Luo et al. (2022) specifies the number of AGD iterations required to obtain the prescribed inner accuracy.

Corollary 14 (Residual-to-oracle certificate) *Suppose Assumption 1 holds and let $M_P = 4\sqrt{2} \kappa^3 \rho$. If*

$$\|\nabla_y f(x, y)\| \leq \tau_{\text{out}}, \quad (38)$$

define

$$\xi(x, y) := \frac{(1 + \kappa)^2 \rho}{\mu} \|\nabla_y f(x, y)\|. \quad (39)$$

Algorithm 1: Known-parameter NC-SC UTR with Newton-corrected derivatives

Input: $x_0, y_{-1}, \epsilon > 0, \ell, \mu, \rho$, and an inner residual module $\mathcal{S}$ satisfying (10)

$\kappa \leftarrow \ell/\mu$

$M_P \leftarrow 4\sqrt{2}\kappa^3\rho$

compute τ_{out} from (37)

$y_0 \leftarrow \text{MaxRes}_{\mathcal{S}}(x_0, y_{-1}, \tau_{\text{out}})$

for $k = 0, 1, \dots$ **do**

$g_k \leftarrow \hat{g}(x_k, y_k), H_k \leftarrow \hat{H}(x_k, y_k)$

$\xi_k \leftarrow (1 + \kappa)^2 \rho \|\nabla_y f(x_k, y_k)\|/\mu$

$h_k \leftarrow \max\{\|g_k\|, \epsilon\}$

if $\|g_k\| \leq \epsilon$ **and** $\lambda_{\min}(H_k) \geq -\sqrt{M_P\epsilon} + \xi_k$ **then**

return x_k

compute a global solution d_k of

$$\min_{\|d\| \leq \sqrt{h_k}/(4\sqrt{M_P/6})} g_k^\top d + \frac{1}{2} d^\top \left(H_k + \xi_k I + 2\sqrt{\frac{M_P}{6}} \sqrt{h_k} I \right) d;$$

$x_{k+1} \leftarrow x_k + d_k$

$y_{k+1} \leftarrow \text{MaxRes}_{\mathcal{S}}(x_{k+1}, y_k, \tau_{\text{out}})$

Then

$$\|\hat{g}(x, y) - \nabla P(x)\| < \frac{\epsilon}{8192}, \quad \|\hat{H}(x, y) - \nabla^2 P(x)\| \leq \xi(x, y) \leq \frac{1}{64} \sqrt{\frac{M_P}{6}} \sqrt{\epsilon}. \quad (40)$$

Hence the fixed-scale inexact-oracle conditions (30) hold for $F = P$ and $M = M_P$.

The proof is deferred to Appendix B.2. The residual target also satisfies

$$\frac{\tau_{\text{out}}}{\mu} = \Theta\left(\sqrt{\frac{\epsilon}{\kappa\rho}}\right) = \Theta\left(\kappa\sqrt{\frac{\epsilon}{M_P}}\right), \quad (41a)$$

$$\tau_{\text{out}} \geq \frac{\ell}{64\sqrt{3}} \sqrt{\frac{\epsilon}{M_P}}. \quad (41b)$$

The second relation follows directly by substituting $M_P = 4\sqrt{2}\kappa^3\rho$ and using $4\kappa^2/(1+\kappa)^2 \geq 1$.

We can now state the first minimax method.

The residual used in the inner stopping condition is also used in ξ_k , so no additional inner-gradient evaluation is required for the Hessian certificate. If the inner solver happens to return a residual smaller than τ_{out} , the curvature test uses the correspondingly smaller certificate.

Algorithm 1 does not evaluate the primal function P and does not require access to $y^*(x)$. Its outer parameters τ_{out} and M_P , however, depend on ℓ, μ , and ρ ; hence the method in this section is a known-parameter construction even if the particular implementation of $\mathcal{S}$ is itself parameter free.

Theorem 15 (Complexity under an inner residual module) *Suppose Assumption 1 holds. Fix (x_0, y_{-1}) and $\epsilon > 0$, and define*

$$\kappa := \frac{\ell}{\mu}, \quad M_P := 4\sqrt{2}\kappa^3\rho, \quad \Delta_P := P(x_0) - P^*.$$

Run Algorithm 1 with any inner module satisfying (10). Then the algorithm returns $\hat{x}$ such that

$$\|\nabla P(\hat{x})\| \leq \frac{8193}{8192} \epsilon, \quad \lambda_{\min}(\nabla^2 P(\hat{x})) \geq -\sqrt{M_P\epsilon}. \quad (42)$$

If K denotes the number of nonterminal outer iterations, then Algorithm 1 solves exactly K trust-region subproblems and forms $K + 1$ corrected-gradient and Schur-complement Hessian estimators. Moreover,

$$K \leq 65\sqrt{\frac{M_P}{6}} \Delta_P \epsilon^{-3/2} + 3 \log_+ \frac{h_0}{\epsilon} + 2, \quad (43)$$

where

$$y_0 := \text{MaxRes}_{\mathcal{S}}(x_0, y_{-1}, \tau_{\text{out}}), \quad h_0 := \max\{\|\widehat{g}(x_0, y_0)\|, \epsilon\}.$$

Consequently,

$$K = O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2} + \log_+ \frac{\|\nabla P(x_0)\|}{\epsilon} + 1\right). \quad (44)$$

Suppose in addition that $\mathcal{S}$ satisfies the accelerated work bound (11). Then the total number N_y of inner-gradient evaluations satisfies

$$\begin{aligned} N_y = & O\left(C_S \sqrt{\kappa} b_S \left[\Delta_P \sqrt{M_P} \epsilon^{-3/2} + \log_+ \frac{\|\nabla P(x_0)\|}{\epsilon} + 1\right]\right) \\ & + O\left(C_S \sqrt{\kappa} \log_+ \frac{\|\nabla_y f(x_0, y_{-1})\|}{\tau_{\text{out}}}\right). \end{aligned} \quad (45)$$

In particular, if

$$C_S = O(1), \quad b_S = O(1 + \log \kappa),$$

then, for fixed problem data and initialization, the leading term in N_y is

$$O\left((1 + \log \kappa) \Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}\right), \quad (46)$$

and no $\log(1/\epsilon)$ factor multiplies this leading term. Replacing the internal tolerance by $8192\epsilon/8193$ gives the standard $(\epsilon, \sqrt{M_P\epsilon})$ guarantee without changing these orders.

The proof is given in Appendix B.3. The outer part follows immediately by combining Corollary 14 with Theorem 12. The nontrivial additional point is the summation of the per-call logarithmic inner costs, which we describe next.

3.3 Total Inner-Gradient Complexity

A direct worst-case estimate of every inner call would obscure the benefit of warm starts. The proof of Theorem 15 instead bounds the total logarithmic work along the trajectory.

Suppose y_k satisfies

$$\|\nabla_y f(x_k, y_k)\| \leq \tau_{\text{out}}.$$

After the outer step $x_{k+1} = x_k + d_k$, joint ℓ -smoothness of f gives

$$\|\nabla_y f(x_{k+1}, y_k)\| \leq \tau_{\text{out}} + \ell \|d_k\|. \quad (47)$$

Hence, apart from the initial call, the logarithmic term in (11) is bounded by

$$1 + \log_2 \left(1 + \frac{\ell \|d_k\|}{\tau_{\text{out}}}\right). \quad (48)$$

By (41b),

$$\frac{\ell \|d_k\|}{\tau_{\text{out}}} \leq 64\sqrt{3} \|d_k\| \sqrt{\frac{M_P}{\epsilon}}. \quad (49)$$

Using

$$\log_2(1+t) = O(1+t^3), \quad t \geq 0,$$

we obtain

$$\log_2\left(1 + \frac{\ell\|d_k\|}{\tau_{\text{out}}}\right) = O\left(1 + \left(\|d_k\|\sqrt{\frac{M_P}{\epsilon}}\right)^3\right). \quad (50)$$

Lemma 13, applied with $F = P$ and $M = M_P$, therefore gives

$$\begin{aligned} \sum_{k=0}^{K-1} \left(\|d_k\|\sqrt{\frac{M_P}{\epsilon}}\right)^3 &= \frac{M_P^{3/2}}{\epsilon^{3/2}} \sum_{k=0}^{K-1} \|d_k\|^3 \\ &\leq 4\Delta_P \sqrt{M_P} \epsilon^{-3/2} + \frac{K}{98\,304\sqrt{6}}. \end{aligned} \quad (51)$$

Thus the per-call logarithms are paid for by the same $\epsilon^{-3/2}$ -scale outer budget, rather than by multiplying the number of outer iterations by a worst-case $\log(1/\epsilon)$ factor.

We next verify that a standard accelerated method realizes the abstract work condition.

For fixed x , consider accelerated gradient descent applied to

$$\phi_x(y) := -f(x, y).$$

Starting from $y^{(0)}$, set $v^{(0)} = y^{(0)}$ and

$$\begin{aligned} y^{(j+1)} &= v^{(j)} + \frac{1}{\ell} \nabla_y f(x, v^{(j)}), \\ v^{(j+1)} &= y^{(j+1)} + \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} (y^{(j+1)} - y^{(j)}). \end{aligned} \quad (52)$$

Proposition 16 (AGD realization of the residual module) *Suppose Assumption 1 holds. Fix x , an initial point $y^{(0)}$, and $\tau > 0$, and define*

$$R_{\text{in}} := \|\nabla_y f(x, y^{(0)})\|.$$

For $\kappa > 1$, the iterates in (52) satisfy

$$\|y^{(N)} - y^*(x)\| \leq \sqrt{\kappa + 1} \left(1 - \frac{1}{\sqrt{\kappa}}\right)^{N/2} \|y^{(0)} - y^*(x)\|. \quad (53)$$

If $R_{\text{in}} \leq \tau$, return $y^{(0)}$. Otherwise, it is sufficient to take

$$N := \left\lceil 2\sqrt{\kappa} \log_+ \left(\frac{\sqrt{\kappa + 1} \kappa R_{\text{in}}}{\tau} \right) \right\rceil \quad (54)$$

and verify the residual at the terminal iterate. Counting the initial gradient once and reusing it in the first AGD step, this requires at most

$$1 + \left\lceil 2\sqrt{\kappa} \log_+ \left(\frac{\sqrt{\kappa + 1} \kappa R_{\text{in}}}{\tau} \right) \right\rceil \quad (55)$$

inner-gradient evaluations. Consequently, AGD satisfies (11) with

$$C_S = O(1), \quad b_S = O(1 + \log \kappa).$$

When $\kappa = 1$, one gradient step is sufficient.

The contraction in (53) is the standard accelerated estimate for a smooth strongly convex objective; the short verification of the residual complexity is given in Appendix B.3.

Corollary 17 (AGD implementation of Algorithm 1) *Choose the inner module in Algorithm 1 to be the AGD implementation in Proposition 16. Then Algorithm 1 has the outer complexity (44) and, in its leading term, uses*

$$O\left((1 + \log \kappa) \Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}\right) \quad (56)$$

inner-gradient evaluations. No $\log(1/\epsilon)$ multiplies this leading term.

Remark 18 (Oracle accounting) *If Algorithm 1 solves K trust-region subproblems, it performs K nonterminal outer iterations and constructs $K + 1$ corrected gradients and Schur-complement Hessians, including the terminal test. The linear algebra required to apply $(\nabla_{yy}^2 f)^{-1}$ is not counted as an inner-gradient evaluation. If these inverse actions are implemented iteratively using Hessian-vector products, their complexity must be accounted for separately. The terminal evaluation of $\nabla_y f$ used to verify the residual may be reused when forming the corrected gradient.*

Remark 19 (Comparison with the raw-gradient analysis) *MCN (Luo et al., 2022) uses the raw plug-in gradient $\nabla_x f(x, y)$, for which*

$$\|\nabla_x f(x, y) - \nabla P(x)\| \leq \ell \|y - y^*(x)\|. \quad (57)$$

Thus an $O(\epsilon)$ gradient-estimation error requires an inner distance of order $O(\epsilon/\ell)$.

More explicitly, after the summation in Luo et al. (2022, Theorem 2), the target-accuracy-dependent logarithm in their bound involves

$$\log\left(\kappa^{3/2} + \frac{\epsilon^{3/2}}{\rho^{3/2} \tilde{\epsilon}^3}\right), \quad \tilde{\epsilon} = \min\left\{\frac{\epsilon}{192\ell}, \frac{\sqrt{M_P \epsilon}}{48\rho}\right\}. \quad (58)$$

In the small- ϵ regime, this logarithm scales as

$$\log\left(\kappa^{3/2} + \left(\frac{\ell}{\sqrt{\rho\epsilon}}\right)^3\right). \quad (59)$$

The known quadratic accuracy of the implicit gradient (Ablin et al., 2020) instead permits an inner distance of order $\sqrt{\epsilon}$. The improvement in our total-work bound combines this accuracy with the cumulative cubic path estimate and the summation of warm-started inner costs. At the level of these two explicit worst-case upper bounds, the multiplicative target-accuracy logarithm is removed while retaining the $1 + \log \kappa$ dependence of the accelerated inner realization.

4 Parameter-Free Second-Order Methods

Algorithm 1 achieves the desired $\epsilon^{-3/2}$ leading outer dependence and a sharper total inner-gradient bound, but its outer scale M_P and residual target τ_{out} depend on the unknown regularity constants ℓ , μ , and ρ . We now remove these quantities from the executable method.

We give two parameter-free constructions. The first combines an adaptive UTR scheme with uniformly accurate inner solves. It retains the $\epsilon^{-3/2}$ leading outer dependence, but its total inner-gradient complexity contains a multiplicative $\log(1/\epsilon)$. The second construction instead tracks the sequence of inner maximizers through persistent SCAR calls and adapts the inner accuracy to the current outer scale. This removes the target-accuracy logarithm from the leading inner-gradient term.

Both results below are class-uniform asymptotic statements. After fixing common constants (ℓ, μ, ρ) , there exists an unknown threshold $\epsilon_0(\ell, \mu, \rho) > 0$ such that the conclusions hold simultaneously for every payoff satisfying Assumption 1 with these constants and every initialization. The algorithms themselves do not use these constants and cannot determine whether a requested tolerance lies below the corresponding threshold.

4.1 A Uniform-Accuracy Parameter-Free Method

We first recall the parameter-free strongly convex solver used for the inner maximization. For fixed x , define

$$q_x(y) := -f(x, y).$$

Under Assumption 1, q_x is μ -strongly convex and has an ℓ -Lipschitz gradient.

SCAR as an inner residual solver. We use the following result of Lan et al. (2026).

Theorem 20 (SCAR; Lan et al. (2026)) *Let $\phi : \mathbb{R}^m \rightarrow \mathbb{R}$ be differentiable, μ -strongly convex, and have an L -Lipschitz gradient, with $0 < \mu < L$. Let $\tau > 0$ and $u_0 \in \mathbb{R}^m$. If $\|\nabla\phi(u_0)\| \leq \tau$, return u_0 . Otherwise, initialize SCAR with*

$$\mu_0 \geq \mu, \quad 0 < M_0 \leq 4L.$$

Then SCAR returns $\hat{u}$ satisfying

$$\|\nabla\phi(\hat{u})\| \leq \tau \tag{60}$$

using at most

$$(4 + 8\sqrt{5}C_1) \sqrt{\frac{L}{\mu}} \left[\left\lceil \log_4 \frac{\mu_0}{\mu} \right\rceil + \left\lceil \log_2 \frac{\|\nabla\phi(u_0)\|}{\tau} \right\rceil \right] \tag{61}$$

gradient evaluations, where C_1 is the universal constant in Theorem 4.1 of Lan et al. (2026). The line-search estimate returned by SCAR is at most $2L$.

The cited theorem is stated for the strict case $\mu < L$. The endpoint $L = \mu$ can be handled separately by the same AR estimate and contributes only a numerical constant; see Appendix C.1.

SCAR can be initialized without knowing either curvature parameter. Given x , an initial point y^{in} , and any $z \neq y^{\text{in}}$, define the observable secant

$$a_x := \frac{\|\nabla_y f(x, z) - \nabla_y f(x, y^{\text{in}})\|}{\|z - y^{\text{in}}\|}. \tag{62}$$

Corollary 21 (Parameter-free SCAR residual module) *Under Assumption 1, run SCAR on $q_x = -f(x, \cdot)$ with*

$$\mu_0 = M_0 = a_x$$

and residual tolerance τ . No value of ℓ , μ , or κ is supplied to the inner solver, and the returned point $\hat{y}$ satisfies

$$\|\nabla_y f(x, \hat{y})\| \leq \tau.$$

Including the secant initialization, the number of inner-gradient evaluations is

$$O\left(\sqrt{\kappa} \left[1 + \log \kappa + \log_+ \frac{\|\nabla_y f(x, y^{\text{in}})\|}{\tau}\right]\right). \tag{63}$$

Hence secant-initialized SCAR realizes the residual interface in Definition 5 without using the problem regularity constants.

The proof is deferred to Appendix C.1.

Uniform inner accuracy. The first parameter-free construction uses the same absolute residual target at every inner call:

$$\tau_{\text{pf}} := \epsilon^2, \quad 0 < \epsilon < 1. \quad (64)$$

The nonmonotone value allowance used below is ϵ^3 .

Although these quantities do not involve unknown regularity parameters, sufficiently small ϵ is still required in the analysis to convert the observable residual into the desired primal-function oracle accuracy.

Lemma 22 (Class-uniform asymptotic oracle accuracy) *Fix $\ell \geq \mu > 0$ and $\rho > 0$, and define*

$$\kappa := \frac{\ell}{\mu}, \quad M_P := 4\sqrt{2} \kappa^3 \rho.$$

There exists $\epsilon_{\text{or}} = \epsilon_{\text{or}}(\ell, \mu, \rho) > 0$ such that, simultaneously for every payoff satisfying Assumption 1 with these common constants and every $0 < \epsilon \leq \epsilon_{\text{or}}$, any point satisfying

$$\|\nabla_y f(x, y)\| \leq \epsilon^2$$

also satisfies

$$\|\hat{g}(x, y) - \nabla P(x)\| \leq \frac{\epsilon}{256}, \quad (65a)$$

$$\|\hat{H}(x, y) - \nabla^2 P(x)\| \leq \frac{1}{64} \sqrt{\frac{M_P \epsilon}{6}}, \quad (65b)$$

$$|\hat{P}(x, y) - P(x)| \leq \frac{\epsilon^3}{8}. \quad (65c)$$

The threshold is used only in the analysis and is not supplied to the algorithm.

Algorithm 2. At a current stored tuple (x, y) satisfying (64), let

$$(\hat{P}, g, H) := (\hat{P}(x, y), \hat{g}(x, y), \hat{H}(x, y)), \quad h := \max\{\|g\|, \epsilon\}.$$

For a trial scale $\sigma \geq \epsilon$, consider

$$\min_{\|d\| \leq \sqrt{h}/(4\sigma)} g^\top d + \frac{1}{2} d^\top \left(H + \frac{\sigma \sqrt{\epsilon}}{64} I + \sigma \sqrt{h} I \right) d. \quad (66)$$

Every quantity tested by Algorithm 2 is observable. In particular, if $\mathcal{S}$ is chosen as the secant-initialized SCAR module in Corollary 21, the executable method uses none of

$$\ell, \quad \mu, \quad \rho, \quad \kappa, \quad M_P, \quad \Delta_P.$$

The fixed inner target ϵ^2 is independent of the trial scale, so rejection changes only the outer trust-region model; the corrected tuple at the current point remains unchanged.

The proof uses the fact that once

$$\sigma^2 \geq \frac{M_P}{6},$$

the Hessian padding is large enough for the inexact trust-region estimate of Lemma 10. A finite-prefix counting argument then controls accepted and rejected trials. The detailed safe-scale and counting lemmas are deferred to Appendix C.1.

Algorithm 2: Uniform-accuracy parameter-free NC-SC UTR

Input: x_0, y_{-1} , $0 < \epsilon < 1$, and a parameter-free inner residual module $\mathcal{S}$ satisfying (10)
 $x \leftarrow x_0$, $\sigma \leftarrow \epsilon$, $\eta \leftarrow 1/256$
 $y \leftarrow \text{MaxRes}_{\mathcal{S}}(x_0, y_{-1}, \epsilon^2)$
compute $(\hat{P}, g, H)$ from (12) and (13)
while *true* **do**
 $h \leftarrow \max\{\|g\|, \epsilon\}$
 if $\|g\| \leq \epsilon$ *and* $\lambda_{\min}(H) \geq -(9/8)\sigma\sqrt{\epsilon}$ **then**
 return x
 solve (66) globally and obtain (d, λ) ; $r \leftarrow \|d\|$
 if $r = 0$ **then**
 return x
 $x^+ \leftarrow x + d$
 $y^+ \leftarrow \text{MaxRes}_{\mathcal{S}}(x^+, y, \epsilon^2)$
 compute $(\hat{P}^+, g^+, H^+)$ at (x^+, y^+)
 if $r < \sqrt{h}/(4\sigma)$, $\hat{P}^+ \leq \hat{P} + \epsilon^3$, *and* $\|g^+\| \leq h/2$ **then**
 accept the trial
 else
 if $r = \sqrt{h}/(4\sigma)$, $\hat{P} - \hat{P}^+ \geq \eta(h^{3/2}/\sigma + \lambda r^2) + \epsilon^3$, *and* $\|g^+\| \leq h/2 + \lambda r$ **then**
 accept the trial
 else
 reject the trial
 if *the trial is accepted* **then**
 $(x, y, \hat{P}, g, H) \leftarrow (x^+, y^+, \hat{P}^+, g^+, H^+)$
 $\sigma \leftarrow \max\{\epsilon, \sigma/2\}$
 else
 discard the candidate tuple
 $\sigma \leftarrow 2\sigma$

Theorem 23 (Uniform-accuracy parameter-free complexity) *Fix $\ell \geq \mu > 0$ and $\rho > 0$. There exists*

$$\epsilon_0 = \epsilon_0(\ell, \mu, \rho) \in (0, 1)$$

such that the following statement holds simultaneously for every payoff satisfying Assumption 1 with these common constants and every initialization (x_0, y_{-1}) .

Define, only for the purpose of stating the bound,

$$\kappa := \frac{\ell}{\mu}, \quad M_P := 4\sqrt{2}\kappa^3\rho, \quad \Delta_P := P(x_0) - P^*.$$

For every $0 < \epsilon \leq \epsilon_0$, Algorithm 2 terminates and returns $\hat{x}$ satisfying

$$\|\nabla P(\hat{x})\| \leq \frac{257}{256}\epsilon, \quad \lambda_{\min}(\nabla^2 P(\hat{x})) \geq -\sqrt{M_P\epsilon}. \quad (67)$$

Consequently, using internal tolerance $256\epsilon/257$ gives the standard $(\epsilon, \sqrt{M_P\epsilon})$ guarantee.

If Q denotes the total number of solved robust trust-region subproblems, then

$$Q = O\left(\Delta_P\sqrt{M_P}\epsilon^{-3/2} + \log_+ \frac{\|\nabla P(x_0)\|}{\epsilon} + \log_+ \frac{\sqrt{M_P}}{\epsilon} + 1\right). \quad (68)$$

Thus its leading outer dependence is

$$O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}\right). \quad (69)$$

This theorem is an asymptotic parameter-free result in the precise class-uniform sense stated above: the algorithm does not use the regularity constants, but the theorem does not provide an observable certificate that a given ϵ lies below $\epsilon_0(\ell, \mu, \rho)$.

We next count the work of the concrete SCAR implementation.

Corollary 24 (Uniform-accuracy SCAR complexity) *Under the hypotheses of Theorem 23, choose $\mathcal{S}$ to be the secant-initialized SCAR module in Corollary 21. Let Q be the number of robust trust-region subproblems and define*

$$G_Q := 1 + \sqrt{2(1 + \kappa)\ell \left(\Delta_P + \frac{5}{4}Q\epsilon^3 \right)}. \quad (70)$$

Then the total number N_y of inner-gradient evaluations satisfies

$$\begin{aligned} N_y = & O\left(\sqrt{\kappa}(Q+1) \left[1 + \log \kappa + \log\left(1 + \frac{\ell\sqrt{G_Q}}{4\epsilon^3}\right)\right]\right) \\ & + O\left(\sqrt{\kappa} \left[1 + \log \kappa + \log_+ \frac{\|\nabla_y f(x_0, y_{-1})\|}{\epsilon^2}\right]\right). \end{aligned} \quad (71)$$

In particular, the uniformly accurate SCAR calls introduce a multiplicative $\log(1/\epsilon)$ in the leading target-accuracy dependence of the total inner-gradient count.

Thus Algorithm 2 already removes all unknown regularity constants from the executable method while retaining the desired leading outer rate. Its remaining weakness is not outer adaptivity, but the decision to solve every candidate inner problem to the same absolute residual ϵ^2 .

4.2 Persistent Inner Tracking

We now replace independent high-accuracy inner solves by tracking the sequence of inner maximizers. The central idea is to retain the curvature calibration learned by SCAR across successive objectives

$$q_x(y) = -f(x, y),$$

while restarting accelerated momentum whenever the outer point changes.

Rather than solving every inner problem independently to a prescribed absolute accuracy, the method maintains only the accuracy required by the current outer scale.

Persistent SCAR. We first record the sequential SCAR property used by the algorithm.

Lemma 25 (Persistent SCAR halving) *Let $\{\phi_j\}_{j \geq 1}$ be any sequence of differentiable functions that are all μ -strongly convex and have L -Lipschitz gradients. Set $\kappa := L/\mu$.*

Suppose the retained strong-convexity guess ν_0 and line-search estimate $\mathcal{M}_0$ satisfy

$$\mu \leq \nu_0 \leq L, \quad 0 < \mathcal{M}_0 \leq 4L.$$

A persistent halving call at (ϕ_j, u) freezes

$$R := \|\nabla \phi_j(u)\|$$

and repeatedly applies the AR call used by SCAR until it obtains v satisfying

$$\|\nabla\phi_j(v)\| \leq \frac{R}{2}.$$

A failed curvature guess replaces ν by $\nu/4$ while retaining the updated line-search estimate.

Across any N successful halving calls, the total number of failed curvature guesses is at most

$$\left\lceil \log_4 \frac{\nu_0}{\mu} \right\rceil \leq \lceil \log_4 \kappa \rceil, \quad (72)$$

and the total number of gradient evaluations is at most

$$(4 + 2\sqrt{10} C_1) \sqrt{\kappa} \left(N + \left\lceil \log_4 \frac{\nu_0}{\mu} \right\rceil \right). \quad (73)$$

After any nontrivial AR call, the retained line-search estimate is at most $2L$.

The crucial distinction from restarted SCAR is that the $\log \kappa$ curvature-calibration cost in (72) is paid across the complete sequence, rather than independently at every outer call.

Working accuracy and tracking. Define

$$\mathcal{A}_\epsilon := \log(e/\epsilon), \quad \sigma_0 := \mathcal{A}_\epsilon^{-1}, \quad c_0 := 2^{-12}, \quad r_\epsilon(\sigma) := \frac{\sqrt{\epsilon}}{4\sigma}. \quad (74)$$

The quantity $r_\epsilon(\sigma)$ is an inner accuracy scale and should not be confused with the trust-region step length.

A stored tuple (x, y) is called *working-accurate* at scale σ if

$$\|\nabla_y f(x, y)\| \leq c_0 \ell r_\epsilon(\sigma). \quad (75)$$

The constant ℓ appears only in the analysis; the executable method maintains this invariant through observable residual halvings.

Lemma 26 (Variable-step tracking) Suppose (75) holds at (x, y) , and let d be an outer step with $r := \|d\|$. Starting from y at $x + d$, apply

$$N_{\text{tr}}(r, \sigma) := \left\lceil \log_2 \left(1 + \frac{r}{c_0 r_\epsilon(\sigma)} \right) \right\rceil \quad (76)$$

successful persistent SCAR halvings. The resulting point satisfies (75) at $x + d$ with the same scale. At an unchanged outer point, one successful halving restores the invariant after σ is doubled.

For later use, set

$$C_g := \frac{c_0^2(1 + \kappa)\rho\kappa^2}{32}, \quad C_v := \frac{7c_0^3\rho\kappa^3}{1536}. \quad (77)$$

Lemma 27 (Working-oracle accuracy) Fix common constants (ℓ, μ, ρ) . For all sufficiently small ϵ , with a threshold depending only on these constants, every working-accurate tuple at $\sigma \geq \sigma_0$ satisfies

$$\|\hat{g} - \nabla P(x)\| \leq C_g \frac{\epsilon}{\sigma^2}, \quad |\hat{P} - P(x)| \leq C_v \frac{\epsilon^{3/2}}{\sigma^3}. \quad (78)$$

If additionally

$$\sigma^2 \geq \frac{M_P}{6},$$

then

$$\begin{aligned}\|\hat{g} - \nabla P(x)\| &\leq \epsilon/1024, \\ \|\hat{H} - \nabla^2 P(x)\| &\leq \sigma\sqrt{\epsilon}/64, \\ |\hat{P} - P(x)| &\leq \epsilon^{3/2}/(4096\sigma).\end{aligned}\tag{79}$$

Furthermore, for any secant a_x of the form (62), any validation point satisfying

$$\|\nabla_y f(x, y_v)\| \leq c_0 a_x \epsilon^{3/2}\tag{80}$$

obeys

$$\begin{aligned}\|\hat{g}(x, y_v) - \nabla P(x)\| &\leq \epsilon/1024, \\ \|\hat{H}(x, y_v) - \nabla^2 P(x)\| &\leq \sigma\sqrt{\epsilon}/32\end{aligned}\tag{81}$$

for every $\sigma \geq \sigma_0$.

All three preceding lemmas are proved in Appendix C.2.

Algorithm 3. Given a working-accurate tuple, define

$$h := \max\{\|\hat{g}\|, \epsilon\}, \quad R(h, \sigma) := \frac{\sqrt{h}}{4\sigma},\tag{82}$$

and solve

$$\min_{\|d\| \leq R(h, \sigma)} \hat{g}^\top d + \frac{1}{2} d^\top \left(\hat{H} + \frac{\sigma\sqrt{\epsilon}}{64} I + \sigma\sqrt{h} I \right) d.\tag{83}$$

For a global solution d and its multiplier $\lambda \geq 0$, write

$$r := \|d\|, \quad W := \sigma\sqrt{h} r^2, \quad U := \lambda r^2, \quad T := \frac{\epsilon^{3/2}}{\sigma}.\tag{84}$$

The scale in Algorithm 3 is nondecreasing. An accepted candidate tuple is stored without further inner refinement. The persistent SCAR state $(\nu, \mathcal{M})$ is retained after accepted trials, rejected trials, and failed validations, while accelerated momentum is restarted whenever the inner objective changes.

Safe scale and accumulated work. At every sufficiently small target tolerance, once

$$\sigma^2 \geq \frac{M_P}{6},$$

the working-oracle errors are small enough for every ordinary trial to satisfy the two acceptance tests. Interior trials with $h = \epsilon$ also pass the high-accuracy validator.

Lemma 28 (Safe scale) For fixed common constants (ℓ, μ, ρ) and all sufficiently small ϵ , if

$$\sigma^2 \geq \frac{M_P}{6},$$

then every ordinary trial of Algorithm 3 is accepted, and every trial with

$$h = \epsilon, \quad r < R(h, \sigma)$$

passes validation.

At any generated scale $\sigma \geq \sigma_0$, successful validation implies

$$\|\nabla P(x^+)\| < \epsilon, \quad \lambda_{\min}(\nabla^2 P(x^+)) \geq -\frac{85}{32}\sigma\sqrt{\epsilon}.\tag{85}$$

Algorithm 3: Parameter-free gradient-scaled TR with persistent SCAR

Input: x_0, y_{-1} and $0 < \epsilon < 1$
 $\mathcal{A}_\epsilon \leftarrow \log(e/\epsilon)$, $\sigma \leftarrow \mathcal{A}_\epsilon^{-1}$, $c_0 \leftarrow 2^{-12}$, $x \leftarrow x_0$
choose $z_0 \neq y_{-1}$ and compute a_0 from (62)
initialize the persistent SCAR state with $\nu_0 = \mathcal{M}_0 = a_0$
at x_0 , apply persistent SCAR halving from y_{-1} until

$$\|\nabla_y f(x_0, y)\| \leq c_0 a_0 r_\epsilon(\sigma);$$

compute $(\hat{P}, \hat{g}, \hat{H})$
while true do
 $h \leftarrow \max\{\|\hat{g}\|, \epsilon\}$
 solve (83) globally and obtain (d, λ)
 $x^+ \leftarrow x + d$, $r \leftarrow \|d\|$
 form (W, U, T) from (84)
 starting from y , apply $N_{\text{tr}}(r, \sigma)$ successful persistent SCAR halvings to q_{x^+} and denote the result by y^+
 if $h = \epsilon$ **and** $r < R(h, \sigma)$ **then**
 choose $z^+ \neq y^+$ and compute the secant a^+ at x^+
 continue persistent SCAR halving at x^+ until

$$\|\nabla_y f(x^+, y_v)\| \leq c_0 a^+ \epsilon^{3/2};$$
 compute $(\hat{g}_v, \hat{H}_v)$
 if $\|\hat{g}_v\| \leq \epsilon/2$ **and** $\lambda_{\min}(\hat{H}_v) \geq -(21/8)\sigma\sqrt{\epsilon}$ **then**
 return x^+
 else
 discard the candidate and validator tuples
 $\sigma \leftarrow 2\sigma$
 apply one successful persistent halving at the unchanged x
 replace y by the returned point and recompute $(\hat{P}, \hat{g}, \hat{H})$
 else
 compute $(\hat{P}^+, \hat{g}^+, \hat{H}^+)$ at (x^+, y^+)
 if $\hat{P} - \hat{P}^+ \geq W/8 + U/4 - T/1024$ **and** $\|\hat{g}^+\| \leq h/2 + \lambda r$ **then**
 $(x, y, \hat{P}, \hat{g}, \hat{H}) \leftarrow (x^+, y^+, \hat{P}^+, \hat{g}^+, \hat{H}^+)$
 else
 discard the candidate tuple
 $\sigma \leftarrow 2\sigma$
 apply one successful persistent halving at the unchanged x
 replace y by the returned point and recompute $(\hat{P}, \hat{g}, \hat{H})$

The total-work argument uses a gradient-level potential. Define

$$\mathcal{L}_\epsilon(h) := \lceil \log_2(h/\epsilon) \rceil, \quad \Phi := \hat{P} + \frac{\epsilon^{3/2}}{512\sigma} \mathcal{L}_\epsilon(h). \quad (86)$$

For every ordinary accepted step, the analysis in Appendix C.2 proves

$$\Phi - \Phi^+ \geq \frac{W}{32} + \frac{T}{1024} + \frac{U}{16}. \quad (87)$$

Interior acceptance reduces the gradient level, whereas the multiplier term U pays for a possible increase after a boundary step.

After accounting for the positive jumps caused by scale refreshes, the same argument yields an explicit budget B_ϵ satisfying

$$B_\epsilon = \Delta_P + O_{\text{fixed}}(\epsilon^{3/2} \mathcal{A}_\epsilon^3) \quad (88)$$

and

$$\sum_{\text{accepted}} \sigma_k^2 \|d_k\|^3 \leq 8B_\epsilon. \quad (89)$$

This is the path budget that amortizes the variable tracking work. Indeed, with

$$z_k := \frac{\sigma_k \|d_k\|}{\sqrt{\epsilon}},$$

the tracking rule satisfies

$$N_{\text{tr},k} = \lceil \log_2(1 + 2^{14} z_k) \rceil \leq 16 + z_k^3.$$

Thus the accumulated tracking cost on accepted steps is controlled by (89), rather than by multiplying the number of outer iterations by an independent logarithmic inner cost.

Theorem 29 (Persistent-SCAR parameter-free complexity) *Fix $\ell \geq \mu > 0$ and $\rho > 0$. There exists*

$$\epsilon_0 = \epsilon_0(\ell, \mu, \rho) \in (0, 1)$$

such that the following holds simultaneously for every payoff satisfying Assumption 1 with these common constants and every initialization (x_0, y_{-1}) .

Set, only for the purpose of stating the bound,

$$\kappa := \frac{\ell}{\mu}, \quad M_P := 4\sqrt{2} \kappa^3 \rho, \quad \Delta_P := P(x_0) - P^*.$$

For every $0 < \epsilon \leq \epsilon_0$, Algorithm 3 terminates while using only ϵ and universal numerical constants. Its output x_{out} satisfies

$$\|\nabla P(x_{\text{out}})\| < \epsilon, \quad \lambda_{\min}(\nabla^2 P(x_{\text{out}})) \geq -\frac{85}{16\sqrt{6}} \sqrt{M_P \epsilon}. \quad (90)$$

If Q denotes the total number of solved trust-region subproblems and N_y the total number of inner-gradient evaluations, including the secant evaluations, then

$$Q = O\left(\Delta_P \sqrt{M_P} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\mathcal{A}_\epsilon^3), \quad (91a)$$

$$N_y = O\left(\Delta_P \sqrt{\kappa M_P} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\sqrt{\kappa} \mathcal{A}_\epsilon^3). \quad (91b)$$

Here O_{fixed} may depend on the fixed payoff, regularity constants, and initialization, but not on ϵ . Equivalently, the leading outer and inner terms are

$$O\left(\Delta_P \kappa^{3/2} \sqrt{\rho} \epsilon^{-3/2}\right), \quad O\left(\Delta_P \kappa^2 \sqrt{\rho} \epsilon^{-3/2}\right). \quad (92)$$

Using internal tolerance $(1536/7225)\epsilon$ gives the standard $(\epsilon, \sqrt{M_P \epsilon})$ guarantee whenever the internal tolerance lies in the stated regime.

The theorem removes the multiplicative target-accuracy logarithm from the leading inner-gradient term. In contrast with Corollary 24, the persistent curvature calibration is shared across the entire sequence of inner objectives, and the variable residual reductions are charged to the path budget (89).

Remark 30 (Scope of the parameter-free guarantee) *The threshold in Theorem 29 is uniform over the class with fixed common constants (ℓ, μ, ρ) and over all initializations, but it is not observable to the algorithm. The theorem therefore does not provide an a posteriori finite-accuracy certificate. The distinction between this class-uniform statement and a guarantee uniform over vanishing strong-concavity moduli is discussed in Appendix ??.*

Remark 31 (Oracle accounting) *The quantity N_y counts evaluations of the inner gradient $\nabla_y f$, which is the oracle counted in the SCAR complexity result. SCAR’s line search also evaluates the inner objective. Under the standard implementation considered by Lan et al. (2026), these function-value evaluations have the same leading order as the gradient count, up to an additive line-search logarithm.*

The corrected-gradient and Schur-complement constructions, including the inverse actions involving $\nabla_{yy}^2 f$, are not included in N_y . If these inverse actions are implemented iteratively, their accuracy and oracle complexity must be accounted for separately.

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Appendix A. Proofs for Section 2

A.1 Proof of the Newton-Corrected Estimator Bounds

Proof of Lemma 6

Corrected value. Fix x , and write

$$C := \nabla_{yy}^2 f(x, y), \quad q := \nabla_y f(x, y), \quad \delta := y - y^*(x).$$

Taylor expansion of the inner gradient gives

$$v := q - C\delta, \quad \|v\| \leq \frac{\rho}{2} \|\delta\|^2. \quad (93)$$

Define

$$\bar{C} := \int_0^1 \nabla_{yy}^2 f(x, y - t\delta) dt.$$

The fundamental theorem of calculus gives $q = \bar{C}\delta$, and hence

$$v = (\bar{C} - C)\delta.$$

Both C and $\bar{C}$ satisfy

$$-\ell I \preceq C, \bar{C} \preceq -\mu I,$$

and therefore

$$\|v\| \leq (\ell - \mu) \|\delta\|. \quad (94)$$

Taylor expansion of the payoff around y gives

$$P(x) = f(x, y) - q^\top \delta + \frac{1}{2} \delta^\top C \delta + R, \quad |R| \leq \frac{\rho}{6} \|\delta\|^3. \quad (95)$$

Using $q = C\delta + v$ and completing the square yields

$$P(x) - \hat{P}(x, y) = R + \frac{1}{2} v^\top C^{-1} v. \quad (96)$$

Strong concavity gives $\|C^{-1}\| \leq 1/\mu$. Combining (93) and (94),

$$\|v\|^2 \leq \frac{\rho}{2} \|\delta\|^2 (\ell - \mu) \|\delta\|.$$

Hence

$$\frac{1}{2} |v^\top C^{-1} v| \leq \frac{(\kappa - 1)\rho}{4} \|\delta\|^3.$$

Together with (95), this gives

$$|\hat{P}(x, y) - P(x)| \leq \frac{(3\kappa - 1)\rho}{12} \|\delta\|^3.$$

The inequality

$$\frac{(3\kappa - 1)\rho}{12} \leq \frac{\kappa\rho}{4}$$

follows from $\kappa \geq 1$.

If $\|\delta\| \leq \mu/\rho$, then (93) gives

$$\frac{1}{2} |v^\top C^{-1} v| \leq \frac{\rho^2}{8\mu} \|\delta\|^4 \leq \frac{\rho}{8} \|\delta\|^3.$$

Combining this with $|R| \leq \rho \|\delta\|^3/6$ proves

$$|\hat{P}(x, y) - P(x)| \leq \frac{7\rho}{24} \|\delta\|^3.$$

Corrected gradient. Let

$$y_t := y^*(x) + t(y - y^*(x)), \quad t \in [0, 1].$$

Since f is twice continuously differentiable, the fundamental theorem of calculus gives

$$\nabla_x f(x, y) - \nabla_x f(x, y^*(x)) = \int_0^1 \nabla_{xy}^2 f(x, y_t)(y - y^*(x)) dt, \quad (97)$$

$$\nabla_y f(x, y) = \int_0^1 \nabla_{yy}^2 f(x, y_t)(y - y^*(x)) dt, \quad (98)$$

where the second identity uses $\nabla_y f(x, y^*(x)) = 0$.

Substituting (97)–(98) into (13a) and using (6a) gives

$$\begin{aligned} \widehat{g}(x, y) - \nabla P(x) &= \int_0^1 \left[\nabla_{xy}^2 f(x, y_t) - \nabla_{xy}^2 f(x, y) \right. \\ &\quad \left. + \nabla_{xy}^2 f(x, y)(\nabla_{yy}^2 f(x, y))^{-1}(\nabla_{yy}^2 f(x, y) - \nabla_{yy}^2 f(x, y_t)) \right] \\ &\quad \times (y - y^*(x)) dt. \end{aligned} \quad (99)$$

This identity displays the cancellation of the linear inner error.

Hessian Lipschitz continuity and

$$\left\| \nabla_{xy}^2 f(x, y)(\nabla_{yy}^2 f(x, y))^{-1} \right\| \leq \kappa$$

give

$$\|\widehat{g}(x, y) - \nabla P(x)\| \leq (1 + \kappa)\rho \int_0^1 (1 - t) dt \|y - y^*(x)\|^2.$$

Therefore

$$\|\widehat{g}(x, y) - \nabla P(x)\| \leq \frac{(1 + \kappa)\rho}{2} \|y - y^*(x)\|^2.$$

Schur-complement Hessian. Retain

$$C := \nabla_{yy}^2 f(x, y)$$

and set

$$B := \nabla_{xy}^2 f(x, y), \quad B_* := \nabla_{xy}^2 f(x, y^*(x)), \quad C_* := \nabla_{yy}^2 f(x, y^*(x)).$$

The inverse identity gives

$$\begin{aligned} \|C^{-1} - C_*^{-1}\| &\leq \frac{\rho}{\mu^2} \|y - y^*(x)\|. \end{aligned} \quad (100)$$

The Schur-complement products satisfy

$$\begin{aligned} BC^{-1}B^\top - B_*C_*^{-1}B_*^\top &= (B - B_*)C^{-1}B^\top \\ &\quad + B_*(C^{-1} - C_*^{-1})B^\top \\ &\quad + B_*C_*^{-1}(B - B_*)^\top. \end{aligned} \quad (101)$$

Hessian Lipschitz continuity, $\|B\|, \|B_*\| \leq \ell$, and $\|C^{-1}\|, \|C_*^{-1}\| \leq 1/\mu$ bound the three terms on the right-hand side by

$$\kappa\rho\|y - y^*(x)\|, \quad \kappa^2\rho\|y - y^*(x)\|, \quad \kappa\rho\|y - y^*(x)\|,$$

respectively. Adding the xx -block error yields

$$\begin{aligned}\|\hat{H}(x, y) - \nabla^2 P(x)\| &\leq \rho\|y - y^*(x)\| + 2\kappa\rho\|y - y^*(x)\| \\ &\quad + \kappa^2\rho\|y - y^*(x)\| \\ &= (1 + \kappa)^2\rho\|y - y^*(x)\|.\end{aligned}$$

This proves the Hessian estimate and completes the proof. ■

A.2 Proof of the Common Inexact Trust-Region Estimate

Proof of Lemma 10 Apply Lemma 9 with

$$B = H + aI.$$

Its stationarity and positive-semidefiniteness conditions give

$$q(d) = -\frac{1}{2}d^\top(H + aI)d - \lambda r^2 \leq -\frac{\lambda}{2}r^2,$$

which proves (29a).

The Taylor bound (23b) and the current endpoint errors imply

$$F(x) - F(x + d) \geq -g^\top d - \frac{1}{2}d^\top H d - \delta_0 r - \frac{\xi}{2}r^2 - \frac{M}{6}r^3.$$

Since

$$-g^\top d - \frac{1}{2}d^\top H d = -q(d) + \frac{a}{2}r^2,$$

the model bound gives

$$F(x) - F(x + d) \geq \frac{a - \xi}{2}r^2 + \frac{\lambda}{2}r^2 - \frac{M}{6}r^3 - \delta_0 r,$$

which proves (29b).

For the candidate gradient, decompose

$$\begin{aligned}g^+ &= [g^+ - \nabla F(x + d)] \\ &\quad + [\nabla F(x + d) - \nabla F(x) - \nabla^2 F(x)d] \\ &\quad + [\nabla F(x) - g] + [\nabla^2 F(x) - H]d + [g + Hd].\end{aligned}$$

The stationarity condition in (25), again with $B = H + aI$, gives

$$g + Hd = -(a + \lambda)d.$$

Taking norms and using (23a), (26), and (28) yields

$$\|g^+\| \leq (a + \xi + \lambda)r + \frac{M}{2}r^2 + \delta_0 + \delta_+,$$

which proves (29c). ■

Appendix B. Proofs for Section 3

B.1 Fixed-Scale Inexact UTR

Let $K \in \mathbb{N}_0 \cup \{\infty\}$ denote the first return index of the procedure (32)–(33), with $K = \infty$ if the procedure never returns. For $k < K$, set

$$r_k := \|d_k\|,$$

and let λ_k be the trust-region multiplier associated with (33). Thus Lemma 9 is applied with

$$B = H_k + \xi_k I + 2\sqrt{\frac{M}{6}}\sqrt{h_k} I, \quad R = \frac{\sqrt{h_k}}{4\sqrt{M/6}}.$$

Lemma 32 (Decrease and clipped-gradient recurrence) *Under the assumptions of Theorem 12, every nonterminal step $k < K$ satisfies*

$$F(x_k) - F(x_{k+1}) \geq \frac{3}{4}\sqrt{\frac{M}{6}}\sqrt{h_k} r_k^2 + \frac{1}{2}\lambda_k r_k^2 - \frac{\epsilon r_k}{8192}. \quad (102)$$

Consequently,

$$\lambda_k = 0 : \quad F(x_k) - F(x_{k+1}) \geq -\frac{\epsilon^{3/2}}{201\,326\,592\sqrt{M/6}}, \quad (103a)$$

$$\lambda_k > 0 : \quad F(x_k) - F(x_{k+1}) \geq \frac{1535}{32768} \frac{h_k^{3/2}}{\sqrt{M/6}} + \frac{1}{2}\lambda_k r_k^2. \quad (103b)$$

Moreover,

$$h_{k+1} \leq \max \left\{ \frac{2849}{4096} h_k + \lambda_k r_k, \epsilon \right\}. \quad (104)$$

Proof Apply Lemma 10 with

$$a = \xi_k + 2\sqrt{\frac{M}{6}}\sqrt{h_k}, \quad R = \frac{\sqrt{h_k}}{4\sqrt{M/6}},$$

and

$$\xi = \xi_k, \quad \delta_0 = \delta_+ = \frac{\epsilon}{8192}.$$

Since $r_k \leq R$,

$$\frac{M}{6} r_k^3 \leq \frac{1}{4}\sqrt{\frac{M}{6}}\sqrt{h_k} r_k^2.$$

The certificate padding cancels the Hessian-error term in (29b), giving

$$F(x_k) - F(x_{k+1}) \geq \frac{3}{4}\sqrt{\frac{M}{6}}\sqrt{h_k} r_k^2 + \frac{1}{2}\lambda_k r_k^2 - \frac{\epsilon r_k}{8192},$$

which proves (102).

If $\lambda_k = 0$, minimizing

$$\frac{3}{4}\sqrt{\frac{M}{6}}\sqrt{\epsilon} r^2 - \frac{\epsilon}{8192} r$$

over $r \geq 0$ gives the minimizer

$$r = \frac{\sqrt{\epsilon}}{12288\sqrt{M/6}}$$

and minimum

$$-\frac{\epsilon^{3/2}}{201\,326\,592\sqrt{M/6}},$$

which proves (103a).

If $\lambda_k > 0$, complementarity gives

$$r_k = \frac{\sqrt{h_k}}{4\sqrt{M/6}}.$$

Hence

$$\begin{aligned} \frac{3}{4}\sqrt{\frac{M}{6}}\sqrt{h_k}r_k^2 - \frac{\epsilon r_k}{8192} \\ \geq \left(\frac{3}{64} - \frac{1}{32768}\right) \frac{h_k^{3/2}}{\sqrt{M/6}} = \frac{1535}{32768} \frac{h_k^{3/2}}{\sqrt{M/6}}, \end{aligned}$$

which, together with the multiplier term, proves (103b).

Finally, the gradient estimate (29c) gives

$$\begin{aligned} \|g_{k+1}\| &\leq 2\sqrt{\frac{M}{6}}\sqrt{h_k}r_k + 2\xi_k r_k + \lambda_k r_k + \frac{M}{2}r_k^2 + \frac{\epsilon}{4096} \\ &\leq \left(\frac{1}{2} + \frac{1}{128} + \frac{3}{16} + \frac{1}{4096}\right) h_k + \lambda_k r_k \\ &= \frac{2849}{4096} h_k + \lambda_k r_k. \end{aligned}$$

Here we used

$$\begin{aligned} 2\sqrt{\frac{M}{6}}\sqrt{h_k}r_k &\leq \frac{1}{2}h_k, & 2\xi_k r_k &\leq \frac{1}{128}h_k, \\ \frac{M}{2}r_k^2 &\leq \frac{3}{16}h_k, & \frac{\epsilon}{4096} &\leq \frac{h_k}{4096}. \end{aligned}$$

Clipping below at ϵ proves (104). ■

The negative right-hand side in (103a) reflects the fact that an inexact interior step need not decrease F . We next account for these small possible increases over a finite prefix.

For any finite prefix of $N \leq K$ nonterminal iterations, define

$$\mathcal{B}_N := \{0 \leq k < N : \lambda_k > 0\}, \quad \mathcal{I}_N := \{0 \leq k < N : \lambda_k = 0\},$$

and, for $k \in \mathcal{B}_N$, set

$$z_k := \frac{\lambda_k r_k}{h_k}. \tag{105}$$

Lemma 33 (Boundary and interior budgets) *Every finite nonterminal prefix satisfies*

$$|\mathcal{B}_N| \leq \frac{32768}{1535} \sqrt{\frac{M}{6}} \Delta_F \epsilon^{-3/2} + \frac{1}{9\,431\,040} |\mathcal{I}_N|, \tag{106a}$$

$$\sum_{k \in \mathcal{B}_N} z_k \leq 8 \sqrt{\frac{M}{6}} \Delta_F \epsilon^{-3/2} + \frac{1}{25\,165\,824} |\mathcal{I}_N|. \tag{106b}$$

Moreover,

$$|\mathcal{I}_N| \leq |\mathcal{B}_N| + 1 + \frac{\log_+(h_0/\epsilon) + \sum_{k \in \mathcal{B}_N} z_k}{\log(4096/2849)}. \tag{107}$$

Proof Summing (103) and using $F(x_N) \geq F^*$ gives

$$\begin{aligned} \Delta_F &\geq \frac{1535\epsilon^{3/2}}{32768\sqrt{M/6}}|\mathcal{B}_N| + \frac{1}{2} \sum_{k \in \mathcal{B}_N} \lambda_k r_k^2 \\ &\quad - \frac{\epsilon^{3/2}}{201\,326\,592\sqrt{M/6}}|\mathcal{I}_N|. \end{aligned}$$

This directly gives (106a). For $k \in \mathcal{B}_N$,

$$\lambda_k r_k^2 = \frac{z_k h_k^{3/2}}{4\sqrt{M/6}} \geq \frac{z_k \epsilon^{3/2}}{4\sqrt{M/6}},$$

which gives (106b).

For $k \in \mathcal{B}_N$, (104) implies

$$\left\lceil \log \frac{h_{k+1}}{h_k} \right\rceil_+ \leq \left\lceil \log \left(\frac{2849}{4096} + z_k \right) \right\rceil_+ \leq z_k.$$

Indeed, the left-hand side vanishes when $z_k \leq 1247/4096$; above this threshold,

$$z - \log \left(\frac{2849}{4096} + z \right)$$

is increasing and positive.

For an index $k \in \mathcal{I}_N$ with $h_{k+1} > \epsilon$, the quantity $\log h_k$ decreases by at least $\log(4096/2849)$. Now suppose $k \in \mathcal{I}_N$ satisfies $h_{k+1} = \epsilon$ and $k+1 < N$. At iteration $k+1$ the gradient part of the stopping test holds, while the procedure has not terminated. Hence

$$\lambda_{\min}(H_{k+1}) < -\sqrt{M}\epsilon + \xi_{k+1}.$$

Consequently,

$$\begin{aligned} \lambda_{\min} &\left(H_{k+1} + \xi_{k+1}I + 2\sqrt{\frac{M}{6}}\sqrt{\epsilon}I \right) \\ &< -\sqrt{M}\epsilon + 2\xi_{k+1} + 2\sqrt{\frac{M}{6}}\sqrt{\epsilon} \\ &\leq \left(2 + \frac{1}{32} - \sqrt{6} \right) \sqrt{\frac{M}{6}}\sqrt{\epsilon} < 0. \end{aligned}$$

The positive-semidefiniteness condition in Lemma 9 therefore forces $\lambda_{k+1} > 0$. Thus every such floor-hitting index is followed by an index in $\mathcal{B}_N$, except possibly the last prefix index. Hence

$$|\{k \in \mathcal{I}_N : h_{k+1} = \epsilon\}| \leq |\mathcal{B}_N| + 1.$$

For the remaining indices in $\mathcal{I}_N$, telescoping $\log h_k$, using $h_N \geq \epsilon$, and charging every positive increase on $\mathcal{B}_N$ to z_k gives

$$|\{k \in \mathcal{I}_N : h_{k+1} > \epsilon\}| \log(4096/2849) \leq \log_+ \frac{h_0}{\epsilon} + \sum_{k \in \mathcal{B}_N} z_k.$$

Adding the two interior counts proves (107). ■

Proof of Theorem 12 Substituting (106) into (107) gives

$$\begin{aligned} & \left[1 - \frac{1}{9\,431\,040} - \frac{1}{25\,165\,824 \log(4096/2849)} \right] |\mathcal{I}_N| \\ & \leq \left(\frac{32768}{1535} + \frac{8}{\log(4096/2849)} \right) \sqrt{\frac{M}{6}} \Delta_F \epsilon^{-3/2} \\ & \quad + 1 + \frac{\log_+(h_0/\epsilon)}{\log(4096/2849)}. \end{aligned}$$

Using

$$\log(4096/2849) > \frac{363}{1000}$$

and

$$1 - \frac{1}{9\,431\,040} - \frac{1}{25\,165\,824 \log(4096/2849)} > \frac{9\,999\,997}{10\,000\,000},$$

together with (106a), gives

$$N = |\mathcal{I}_N| + |\mathcal{B}_N| \leq 65 \sqrt{\frac{M}{6}} \Delta_F \epsilon^{-3/2} + 3 \log_+ \frac{h_0}{\epsilon} + 2. \quad (108)$$

This estimate is uniform over every finite nonterminal prefix.

If $K = \infty$, then (108) would bound every finite $N \leq K$ by the same constant, which is impossible for arbitrarily large N . Therefore $K < \infty$. Taking $N = K$ proves (34).

At termination, (30) gives

$$\|\nabla F(\hat{x})\| \leq \frac{8193}{8192} \epsilon.$$

By Weyl's inequality and (32),

$$\lambda_{\min}(\nabla^2 F(\hat{x})) \geq \lambda_{\min}(H_K) - \xi_K \geq -\sqrt{M\epsilon}.$$

This proves (35). ■

Proof of Lemma 13 Since

$$r_k \leq \frac{\sqrt{h_k}}{4\sqrt{M/6}},$$

we have

$$\sqrt{h_k} \geq 4\sqrt{\frac{M}{6}} r_k.$$

Equation (102) therefore implies

$$F(x_k) - F(x_{k+1}) \geq \frac{M}{2} r_k^3 - \frac{\epsilon r_k}{8192}.$$

Split the cubic term equally and maximize

$$\frac{\epsilon r}{8192} - \frac{M}{4} r^3$$

over $r \geq 0$. Its maximum is

$$\frac{\epsilon^{3/2}}{393\,216\sqrt{6M}}.$$

Hence

$$F(x_k) - F(x_{k+1}) \geq \frac{M}{4} r_k^3 - \frac{\epsilon^{3/2}}{393\,216\sqrt{6M}}.$$

Summing from $k = 0$ to $K - 1$ and using $F(x_K) \geq F^*$ gives

$$M \sum_{k=0}^{K-1} r_k^3 \leq 4\Delta_F + \frac{K}{98\,304\sqrt{6}} \frac{\epsilon^{3/2}}{\sqrt{M}},$$

which is (36). ■

B.2 Oracle Calibration for the NC–SC Method

Proof of Corollary 14 Lemma 4 gives

$$\|y - y^*(x)\| \leq \frac{\|\nabla_y f(x, y)\|}{\mu}.$$

Hence (14c) gives

$$\|\hat{H}(x, y) - \nabla^2 P(x)\| \leq \xi(x, y).$$

Using $\|\nabla_y f(x, y)\| \leq \tau_{\text{out}}$ and (37),

$$\|y - y^*(x)\| \leq \frac{\sqrt{M_P \epsilon / 6}}{64(1 + \kappa)^2 \rho}.$$

Substitution into (14c) gives

$$\xi(x, y) \leq \frac{1}{64} \sqrt{\frac{M_P}{6}} \sqrt{\epsilon}.$$

Similarly, (14b) yields

$$\begin{aligned} \|\hat{g}(x, y) - \nabla P(x)\| &\leq \frac{M_P \epsilon}{49152(1 + \kappa)^3 \rho} \\ &= \frac{\sqrt{2}}{12288} \left(\frac{\kappa}{1 + \kappa} \right)^3 \epsilon < \frac{\epsilon}{8192}. \end{aligned}$$

This proves (40). ■

B.3 Inner Solver and Complexity of Algorithm 1

AGD REALIZATION

Proof of Proposition 16 Apply the standard accelerated-gradient estimate to the ℓ -smooth and μ -strongly convex function

$$\phi_x(y) := -f(x, y).$$

The estimate (53) follows, for example, from Wang and Li (2020, Lemma 2) or Luo et al. (2022, Lemma 5).

By Lemma 4,

$$\|y^{(0)} - y^*(x)\| \leq \frac{R_{\text{in}}}{\mu}.$$

It is sufficient to obtain

$$\|y^{(N)} - y^*(x)\| \leq \frac{\tau}{\ell},$$

because ℓ -smoothness then gives

$$\|\nabla_y f(x, y^{(N)})\| \leq \tau.$$

Combining these inequalities with

$$1 - \frac{1}{\sqrt{\kappa}} \leq \exp\left(-\frac{1}{\sqrt{\kappa}}\right)$$

gives (54) and (55). For $\kappa = 1$, one gradient step is sufficient. ■

COMPLEXITY THEOREM

Proof of Theorem 15 Let K denote the number of nonterminal outer iterations and set

$$r_k := \|d_k\|.$$

By the residual contract and Corollary 14,

$$\|g_k - \nabla P(x_k)\| \leq \frac{\epsilon}{8192},$$

and

$$\|H_k - \nabla^2 P(x_k)\| \leq \xi_k \leq \frac{1}{64} \sqrt{\frac{M_P}{6}} \sqrt{\epsilon}.$$

Hence the outer sequence generated by Algorithm 1 satisfies the fixed-scale procedure of Section 3.1 with

$$F = P, \quad M = M_P.$$

Theorem 12 therefore gives (42) and (43).

The logarithmic term in (43) contains

$$h_0 = \max\{\|g_0\|, \epsilon\}.$$

Since

$$\|g_0\| \leq \|\nabla P(x_0)\| + \frac{\epsilon}{8192},$$

we have

$$\log_+ \frac{h_0}{\epsilon} = O\left(\log_+ \frac{\|\nabla P(x_0)\|}{\epsilon} + 1\right).$$

Together with

$$\sqrt{M_P} = \sqrt{4\sqrt{2}} \kappa^{3/2} \sqrt{\rho},$$

this proves (44).

It remains to bound the total work of the inner module. At the start of the inner call for x_{k+1} , joint ℓ -smoothness and the previous residual guarantee give

$$\|\nabla_y f(x_{k+1}, y_k)\| \leq \|\nabla_y f(x_k, y_k)\| + \ell \|x_{k+1} - x_k\| \leq \tau_{\text{out}} + \ell r_k.$$

Therefore, apart from the initial call, the logarithmic term in (11) is at most

$$1 + \log_2 \left(1 + \frac{\ell r_k}{\tau_{\text{out}}}\right).$$

By (41b),

$$\frac{\ell r_k}{\tau_{\text{out}}} \leq 64\sqrt{3} r_k \sqrt{\frac{M_P}{\epsilon}}.$$

Since

$$\log_2(1+t) = O(1+t^3), \quad t \geq 0,$$

the logarithmic contribution of each noninitial call is

$$O\left(1 + \left(r_k \sqrt{\frac{M_P}{\epsilon}}\right)^3\right).$$

Applying Lemma 13 with $F = P$ and $M = M_P$ gives

$$\begin{aligned} \sum_{k=0}^{K-1} \left(r_k \sqrt{\frac{M_P}{\epsilon}}\right)^3 &= \frac{M_P^{3/2}}{\epsilon^{3/2}} \sum_{k=0}^{K-1} r_k^3 \\ &\leq 4\Delta_P \sqrt{M_P} \epsilon^{-3/2} + \frac{K}{98304\sqrt{6}}. \end{aligned}$$

Summing the module work bounds over all calls, adding the initial call

$$O\left(C_S \sqrt{\kappa} \left[b_S + \log_+ \frac{\|\nabla_y f(x_0, y_{-1})\|}{\tau_{\text{out}}}\right]\right),$$

and substituting the outer bound for K gives (45).

Finally, if

$$C_S = O(1), \quad b_S = O(1 + \log \kappa),$$

then

$$\sqrt{\kappa} \sqrt{M_P} = \Theta(\kappa^2 \sqrt{\rho}),$$

which gives (46). ■

Appendix C. Proofs for Section 4

C.1 Uniform-Accuracy Parameter-Free Method

Proof of Corollary 21 For $q_x = -f(x, \cdot)$, strong monotonicity and Lipschitz continuity of ∇q_x give

$$\mu \|z - y^{\text{in}}\| \leq \|\nabla q_x(z) - \nabla q_x(y^{\text{in}})\| \leq \ell \|z - y^{\text{in}}\|.$$

Hence

$$\mu \leq a_x \leq \ell.$$

For $\kappa > 1$, the initialization $\mu_0 = M_0 = a_x$ therefore satisfies the hypotheses of Theorem 20, and

$$\log_4 \frac{a_x}{\mu} \leq \log_4 \kappa.$$

Substitution into (61) proves (63).

When $\kappa = 1$, strong monotonicity and Lipschitz continuity with the same modulus imply the corresponding endpoint estimate directly, so the cost of a residual-reduction call is bounded by a numerical constant. ■

Proof of Lemma 22 Let

$$s := \sqrt{M_P/6}.$$

Choose

$$\epsilon_{\text{or}} := \min \left\{ 1, \frac{\mu}{\sqrt{\rho}}, \left(\frac{\mu^2}{128(1+\kappa)\rho} \right)^{1/3}, \left(\frac{s\mu}{64(1+\kappa)^2\rho} \right)^{2/3}, \left(\frac{3\mu^3}{7\rho} \right)^{1/3} \right\}. \quad (109)$$

By Lemma 4,

$$\|y - y^*(x)\| \leq \frac{\epsilon^2}{\mu}.$$

The choice (109) ensures

$$\frac{\epsilon^2}{\mu} \leq \frac{\mu}{\rho}.$$

Lemma 6 therefore gives

$$\begin{aligned} \|\hat{g} - \nabla P\| &\leq \frac{(1+\kappa)\rho}{2\mu^2} \epsilon^4, \\ \|\hat{H} - \nabla^2 P\| &\leq \frac{(1+\kappa)^2\rho}{\mu} \epsilon^2, \\ |\hat{P} - P| &\leq \frac{7\rho}{24\mu^3} \epsilon^6. \end{aligned}$$

The remaining conditions in (109) make these quantities at most

$$\frac{\epsilon}{256}, \quad \frac{s\sqrt{\epsilon}}{64}, \quad \frac{\epsilon^3}{8},$$

respectively. All coefficients depend only on (ℓ, μ, ρ) , so the same threshold applies uniformly over the entire parameter class. $\blacksquare$

Lemma 34 (Safe-scale robust trial) Consider a nonzero trial of Algorithm 2, and write

$$h := \max\{\|g\|, \epsilon\}, \quad r := \|d\|.$$

Suppose both current and candidate inner points satisfy $\|\nabla_y f\| \leq \epsilon^2$, and suppose (65) holds. If

$$\sigma^2 \geq M_P/6,$$

then

$$P(x) - P(x+d) \geq \frac{1}{4}\sigma\sqrt{h}r^2 + \frac{1}{2}\lambda r^2 - \frac{(1+\kappa)\rho}{2\mu^2}\epsilon^4 r, \quad (110a)$$

$$\|g^+\| \leq \frac{117}{256}h + \lambda r. \quad (110b)$$

If in addition

$$r = \frac{\sqrt{h}}{4\sigma},$$

then

$$P(x) - P(x+d) \geq \frac{h^{3/2}}{128\sigma} + \frac{1}{4}\lambda r^2. \quad (111)$$

Proof Apply Lemma 10 with

$$F = P, \quad M = M_P,$$

and

$$\begin{aligned} a &= \sigma\sqrt{h} + \frac{\sigma\sqrt{\epsilon}}{64}, \quad R = \frac{\sqrt{h}}{4\sigma}, \\ \xi &= \frac{1}{64}\sqrt{\frac{M_P\epsilon}{6}}, \quad \delta_0 = \delta_+ = \frac{(1+\kappa)\rho}{2\mu^2}\epsilon^4. \end{aligned}$$

At a safe scale,

$$\xi \leq \frac{\sigma\sqrt{\epsilon}}{64},$$

so

$$a - \xi \geq \sigma\sqrt{h}, \quad a + \xi \leq \sigma\sqrt{h} + \frac{2\sigma\sqrt{\epsilon}}{64}.$$

The trust-region radius and $M_P \leq 6\sigma^2$ imply

$$\frac{M_P}{6}r^3 \leq \frac{1}{4}\sigma\sqrt{h}r^2.$$

Equation (29b) therefore proves (110a).

If the radius is active,

$$\sigma\sqrt{h}r^2 = \frac{h^{3/2}}{16\sigma}.$$

By (65a), the last term in (110a) is at most $h^{3/2}/(1024\sigma)$, which yields (111).

Finally, (29c) gives

$$\|g^+\| \leq \sigma\sqrt{h}r + \lambda r + 2\frac{\sigma\sqrt{\epsilon}}{64}r + \frac{(1+\kappa)\rho}{\mu^2}\epsilon^4 + \frac{M_P}{2}r^2.$$

The four terms other than λr are bounded by

$$\frac{h}{4}, \quad \frac{h}{128}, \quad \frac{h}{128}, \quad \frac{3h}{16},$$

respectively. Their sum is smaller than $117h/256$, which proves (110b). ■

Lemma 35 (Accepted-step accounting) *Consider any finite prefix of N accepted nonterminal steps of Algorithm 2. Suppose (65) holds at every stored tuple and all trial scales are bounded by $\bar{\sigma}$. If*

$$\frac{5\bar{\sigma}\epsilon^{3/2}}{4\eta} \left(1 + \frac{4}{\log 2}\right) \leq \frac{1}{2}, \tag{112}$$

then

$$N = O\left(\bar{\sigma}\Delta_P\epsilon^{-3/2} + \log_+ \frac{\|g_0\|}{\epsilon} + 1\right). \tag{113}$$

Proof For an accepted step k , write

$$h_k := \max\{\|g_k\|, \epsilon\}, \quad r_k := \|d_k\|,$$

and define

$$\mathcal{B}_N := \left\{0 \leq k < N : r_k = \frac{\sqrt{h_k}}{4\sigma_k}\right\},$$

$$\mathcal{I}_N := \left\{ 0 \leq k < N : r_k < \frac{\sqrt{h_k}}{4\sigma_k} \right\}.$$

Every $k \in \mathcal{B}_N$ gives a true decrease of at least

$$\eta \left(\frac{h_k^{3/2}}{\sigma_k} + \lambda_k r_k^2 \right),$$

because the two corrected-value errors sum to at most $\epsilon^3/4$. Every $k \in \mathcal{I}_N$ can increase the true primal value by at most $5\epsilon^3/4$. Telescoping and using $h_k \geq \epsilon$ and $\sigma_k \leq \bar{\sigma}$ give

$$|\mathcal{B}_N| \leq \frac{\bar{\sigma}\Delta_P}{\eta\epsilon^{3/2}} + \frac{5\bar{\sigma}\epsilon^{3/2}}{4\eta} |\mathcal{I}_N|, \quad (114a)$$

$$\sum_{k \in \mathcal{B}_N} \frac{\lambda_k r_k}{h_k} \leq \frac{4\bar{\sigma}\Delta_P}{\eta\epsilon^{3/2}} + \frac{5\bar{\sigma}\epsilon^{3/2}}{\eta} |\mathcal{I}_N|. \quad (114b)$$

For the second inequality we used

$$\lambda_k r_k^2 = \frac{\lambda_k r_k}{h_k} \frac{h_k^{3/2}}{4\sigma_k}.$$

The accepted gradient tests imply

$$h_{k+1} \leq \begin{cases} \max\{h_k/2, \epsilon\}, & k \in \mathcal{I}_N, \\ \max\{h_k/2 + \lambda_k r_k, \epsilon\}, & k \in \mathcal{B}_N. \end{cases}$$

Thus every interior step that does not hit the floor decreases $\log(h_k/\epsilon)$ by at least $\log 2$, while a boundary step can increase it by at most $\lambda_k r_k/h_k$.

Suppose an interior step hits the floor and $k+1 < N$. At the next accepted outer iteration, $h = \epsilon$ and the stopping test fails through its Hessian branch. With

$$A := H + \frac{\sigma\sqrt{\epsilon}}{64}I + \sigma\sqrt{\epsilon}I,$$

failure of the stopping test gives

$$-\lambda_{\min}(A) > \frac{7}{64}\sigma\sqrt{\epsilon}.$$

Lemma 9 therefore forces $\lambda > 0$, and hence the next accepted step lies on the boundary. Thus every nonterminal floor event is injectively followed by an index in $\mathcal{B}_N$, apart from a possible final prefix index. Therefore

$$|\mathcal{I}_N| \leq |\mathcal{B}_N| + 1 + \frac{\log_+(\|g_0\|/\epsilon) + \sum_{k \in \mathcal{B}_N} \lambda_k r_k/h_k}{\log 2}. \quad (115)$$

Substituting (114) and using (112) proves (113). ■

Proof of Theorem 23 Set

$$s := \sqrt{M_P/6}, \quad \bar{\sigma} := 2s.$$

Choose the class-uniform threshold no larger than ϵ_{or} and so that

$$\begin{aligned} \epsilon &\leq s, \\ \frac{5\bar{\sigma}\epsilon^3}{4\eta\epsilon^{3/2}} \left(1 + \frac{4}{\log 2} \right) &\leq \frac{1}{2}, \\ \frac{1}{s\sqrt{\epsilon}} \left(\frac{(1+\kappa)\rho}{2\mu^2} \epsilon^4 \right)^2 &\leq \frac{3}{4}\epsilon^3, \\ \left(\frac{1}{128} - \eta \right) \frac{\epsilon^{3/2}}{\bar{\sigma}} &\geq \frac{5}{4}\epsilon^3. \end{aligned} \quad (116)$$

For example, the choice

$$\epsilon_0 := \frac{1}{2} \min \left\{ \epsilon_{\text{or}}, s, \left(\frac{\eta}{5s(1+4/\log 2)} \right)^{2/3}, \left(\frac{3s\mu^4}{(1+\kappa)^2\rho^2} \right)^{2/9} \right\} \quad (117)$$

is sufficient. This threshold depends only on (ℓ, μ, ρ) .

Starting from $\sigma = \epsilon$, a rejected trial doubles σ . Unless the method has already terminated, the first nonzero trial with $\sigma \geq s$ satisfies

$$\sigma \leq 2s = \bar{\sigma}.$$

Lemma 34 implies that its gradient acceptance test holds.

For an interior trial, minimizing the right-hand side of (110a) over $r \geq 0$ shows that the possible true increase is at most

$$\frac{1}{\sigma\sqrt{h}} \left(\frac{(1+\kappa)\rho}{2\mu^2} \epsilon^4 \right)^2.$$

By (116), this is at most $3\epsilon^3/4$. After adding the two corrected-value errors, the interior value test holds.

On a boundary trial, (111) and the two value errors give

$$\hat{P} - \hat{P}^+ \geq \frac{h^{3/2}}{128\sigma} + \frac{1}{4}\lambda r^2 - \frac{\epsilon^3}{4}.$$

The last condition in (116) implies the boundary acceptance test. Hence a safe-scale trial either terminates or is accepted, and no generated scale exceeds $\bar{\sigma}$.

Consider a finite trial prefix with K_N accepted and U_N rejected nonterminal trials. Lemma 35 bounds K_N independently of the prefix length. Because each rejection doubles σ and each acceptance decreases it by at most a factor two,

$$U_N \leq K_N + \log_2(\bar{\sigma}/\epsilon) + 1.$$

Hence every finite prefix has a common finite length bound, so an infinite run is impossible.

For the complete run,

$$Q \leq K + U + 1.$$

Moreover,

$$\|g_0\| \leq \|\nabla P(x_0)\| + \frac{\epsilon}{256},$$

so the initial logarithm can be replaced by $\log_+(\|\nabla P(x_0)\|/\epsilon)$ up to a universal additive constant. This proves (68).

If the explicit stopping test returns, then (65a), (65b), and $\sigma \leq 2s$ give

$$\|\nabla P(x)\| \leq \frac{257}{256}\epsilon,$$

and

$$\lambda_{\min}(\nabla^2 P(x)) \geq -\frac{145}{64}s\sqrt{\epsilon} \geq -\sqrt{M_P\epsilon}.$$

If the robust subproblem returns $d = 0$, global optimality gives $g = 0$, $h = \epsilon$, and

$$H + \frac{\sigma\sqrt{\epsilon}}{64}I + \sigma\sqrt{\epsilon}I \succeq 0.$$

The same oracle bounds then imply

$$\|\nabla P(x)\| \leq \frac{\epsilon}{256}, \quad \lambda_{\min}(\nabla^2 P(x)) \geq -\frac{131}{64}s\sqrt{\epsilon} \geq -\sqrt{M_P\epsilon}.$$

This proves (67). ■

Proof of Corollary 24 Every accepted boundary step decreases the true primal value, while every accepted interior step can increase it by at most $5\epsilon^3/4$. Since there are at most Q accepted steps,

$$P(x) - P^* \leq \Delta_P + \frac{5}{4}Q\epsilon^3$$

at every accepted outer point.

Because ∇P is $(1 + \kappa)\ell$ -Lipschitz, the standard descent inequality together with (65a) gives

$$h \leq G_Q.$$

At a candidate point, SCAR is warm-started from a point whose previous inner residual is at most ϵ^2 . Joint ℓ -smoothness and the trust-region radius give

$$\|\nabla_y f(x + d, y)\| \leq \epsilon^2 + \ell\|d\| \leq \epsilon^2 + \frac{\ell\sqrt{G_Q}}{4\epsilon}.$$

Apply Corollary 21 with target ϵ^2 to every candidate call and separately to the initial call. There is at most one candidate call per solved subproblem, and summing the resulting bounds gives (71). ■

C.2 Persistent Inner Tracking

Proof of Lemma 25 The AR estimate used in the proof of Lan et al. (2026, Theorem 5.1), obtained from their Theorem 4.1, states that a call with current guess ν and an admissible line-search estimate uses at most

$$4 + C_1\sqrt{10L/\nu}$$

gradient evaluations and returns v satisfying

$$\|\nabla\phi_j(v)\| \leq \frac{\nu}{2}\|u - u_j^*\|, \quad u_j^* := \arg \min_u \phi_j(u).$$

If $\nu \leq \mu$, strong convexity gives

$$\|u - u_j^*\| \leq \frac{\|\nabla\phi_j(u)\|}{\mu},$$

so the returned residual is at most $R/2$. Therefore a failed halving implies $\nu > \mu$.

Every failure divides ν by four. Once $\nu \leq \mu$, no later objective in the sequence can cause another curvature-guess failure. Thus the total number of failures is at most

$$\left\lceil \log_4 \frac{\nu_0}{\mu} \right\rceil.$$

Throughout the process,

$$\nu \geq \mu/4,$$

so every attempted stage costs at most

$$4 + C_1\sqrt{40\kappa} \leq (4 + 2\sqrt{10}C_1)\sqrt{\kappa}.$$

There are N successful attempts plus at most the displayed number of failed attempts, which proves (73). The line-search estimate follows from the corresponding SCAR bound. ■

Proof of Lemma 26 Joint Lipschitz continuity of the gradient gives

$$\|\nabla_y f(x+d, y)\| \leq \|\nabla_y f(x, y)\| + \ell r \leq \ell(c_0 r_\epsilon(\sigma) + r).$$

After $N_{\text{tr}}(r, \sigma)$ successful residual halvings, the right-hand side is at most

$$c_0 \ell r_\epsilon(\sigma),$$

which proves the tracking claim. At an unchanged x ,

$$r_\epsilon(2\sigma) = \frac{1}{2} r_\epsilon(\sigma),$$

so one successful halving restores the invariant after scale doubling. ■

Proof of Lemma 27 From strong concavity and (75),

$$e := \|y - y^*(x)\| \leq \frac{c_0 \kappa \sqrt{\epsilon}}{4\sigma} \leq \frac{c_0 \kappa}{4} \sqrt{\epsilon} \mathcal{A}_\epsilon.$$

For all sufficiently small ϵ , uniformly over the fixed parameter class,

$$e \leq \frac{\mu}{\rho}.$$

Lemma 6 therefore gives

$$\begin{aligned} \|\hat{g} - \nabla P(x)\| &\leq \frac{(1+\kappa)\rho}{2} e^2, \\ |\hat{P} - P(x)| &\leq \frac{7\rho}{24} e^3, \end{aligned}$$

and

$$\|\hat{H} - \nabla^2 P(x)\| \leq (1+\kappa)^2 \rho e.$$

Substitution gives (78).

At a safe scale,

$$M_P \leq 6\sigma^2.$$

Using $M_P = 4\sqrt{2}\kappa^3\rho$ and $\kappa \geq 1$ gives

$$\begin{aligned} \|\hat{g} - \nabla P(x)\| &\leq \frac{3c_0^2}{32\sqrt{2}} \epsilon, \\ \|\hat{H} - \nabla^2 P(x)\| &\leq \frac{3\sqrt{2}c_0}{4} \sigma \sqrt{\epsilon}, \\ |\hat{P} - P(x)| &\leq \frac{7c_0^3}{1024\sqrt{2}} \frac{\epsilon^{3/2}}{\sigma}. \end{aligned}$$

Since $c_0 = 2^{-12}$, these imply (79).

For a validation point, $\mu \leq a_x \leq \ell$ and Lemma 4 imply

$$\|y_v - y^*(x)\| \leq c_0 \kappa \epsilon^{3/2}.$$

The corrected-gradient and Hessian errors are therefore at most

$$\frac{(1+\kappa)\rho c_0^2 \kappa^2}{2} \epsilon^3$$

and

$$(1 + \kappa)^2 \rho c_0 \kappa \epsilon^{3/2},$$

respectively. The first is at most $\epsilon/1024$ for sufficiently small ϵ . Since

$$\epsilon \mathcal{A}_\epsilon \rightarrow 0$$

and $\sigma \geq \mathcal{A}_\epsilon^{-1}$, the second is at most $\sigma\sqrt{\epsilon}/32$ for sufficiently small ϵ . This proves (81). ■

Proof of Lemma 28 Apply Lemma 10 with

$$F = P, \quad M = M_P,$$

$$a = \sigma\sqrt{h} + \frac{\sigma\sqrt{\epsilon}}{64}, \quad \xi = \frac{\sigma\sqrt{\epsilon}}{64},$$

and

$$\delta_0 = \delta_+ = \frac{\epsilon}{1024}.$$

The endpoint errors follow from (79). Because

$$r \leq \frac{\sqrt{h}}{4\sigma}, \quad M_P \leq 6\sigma^2,$$

we have

$$\frac{M_P}{6} r^3 \leq \frac{W}{4}.$$

Thus

$$P(x) - P(x+d) \geq \frac{W}{4} + \frac{U}{2} - \frac{\epsilon r}{1024}.$$

Young's inequality and $h \geq \epsilon$ give

$$\frac{\epsilon r}{1024} \leq \frac{W}{8} + \frac{2T}{1024^2}.$$

After paying the two corrected-value errors,

$$\hat{P} - \hat{P}^+ \geq \frac{W}{8} + \frac{U}{2} - \left(\frac{2}{1024^2} + \frac{2}{4096} \right) T \geq \frac{W}{8} + \frac{U}{4} - \frac{T}{1024}.$$

The candidate-gradient estimate similarly gives

$$\|\hat{g}^+\| \leq \lambda r + \frac{h}{4} + \frac{h}{128} + \frac{3h}{16} + \frac{h}{512} = \lambda r + \frac{229}{512}h \leq \lambda r + \frac{h}{2}.$$

Hence every ordinary trial is accepted at a safe scale.

If $h = \epsilon$ and $r < R(h, \sigma)$, then complementarity gives $\lambda = 0$. The candidate-gradient estimate yields

$$\|\nabla P(x+d)\| \leq \frac{457}{1024}\epsilon.$$

Positive semidefiniteness of the trust-region model, the working Hessian error, and Hessian Lipschitz continuity give

$$\lambda_{\min}(\nabla^2 P(x+d)) \geq -\frac{81}{32}\sigma\sqrt{\epsilon}.$$

Using the validation errors,

$$\|\hat{g}_v\| \leq \frac{458}{1024}\epsilon,$$

and

$$\lambda_{\min}(\hat{H}_v) \geq -\frac{82}{32}\sigma\sqrt{\epsilon},$$

so validation succeeds.

Finally, successful validation at any generated scale gives

$$\|\nabla P(x^+)\| < \epsilon$$

and

$$\lambda_{\min}(\nabla^2 P(x^+)) \geq -\left(\frac{21}{8} + \frac{1}{32}\right)\sigma\sqrt{\epsilon} = -\frac{85}{32}\sigma\sqrt{\epsilon}.$$

■

Lemma 36 (Accepted-step potential decrease) *Every ordinary accepted step satisfies*

$$\Phi - \Phi^+ \geq \frac{W}{32} + \frac{T}{1024} + \frac{U}{16}. \quad (118)$$

Proof If $h > \epsilon$ and

$$h^+ \leq \max\{h/2, \epsilon\},$$

then

$$\mathcal{L}_\epsilon(h^+) \leq \mathcal{L}_\epsilon(h) - 1. \quad (119)$$

More generally, if $c \geq 1$ and $h^+ \leq ch$, then

$$\mathcal{L}_\epsilon(h^+) - \mathcal{L}_\epsilon(h) \leq \lceil \log_2 c \rceil \leq 1 + \log_2 c. \quad (120)$$

For an interior accepted step, $\lambda = 0$ and $U = 0$. The gradient acceptance test and (119) give

$$\Phi - \Phi^+ \geq \frac{W}{8} - \frac{T}{1024} + \frac{T}{512} = \frac{W}{8} + \frac{T}{1024}.$$

For a boundary step, let

$$z := \frac{\lambda r}{h}.$$

The gradient acceptance test gives

$$\frac{h^+}{h} \leq \max\{1, 1/2 + z\}.$$

Hence

$$\mathcal{L}_\epsilon(h^+) - \mathcal{L}_\epsilon(h) \leq 1 + \frac{z}{\ln 2}.$$

Since

$$r = \frac{\sqrt{h}}{4\sigma},$$

we have

$$\begin{aligned} W &= \frac{h^{3/2}}{16\sigma} \geq \frac{T}{16}, \\ U &= \frac{zh^{3/2}}{4\sigma}, \quad Tz \leq 4U. \end{aligned}$$

Therefore

$$\begin{aligned}\Phi - \Phi^+ &\geq \frac{W}{8} + \frac{U}{4} - \frac{T}{1024} - \frac{T}{512} \left(1 + \frac{z}{\ln 2}\right) \\ &\geq \frac{5W}{64} + \left(\frac{1}{4} - \frac{1}{128 \ln 2}\right) U.\end{aligned}$$

Finally,

$$\frac{5W}{64} \geq \frac{W}{32} + \frac{T}{1024},$$

and

$$\frac{1}{4} - \frac{1}{128 \ln 2} \geq \frac{1}{16},$$

which proves the claim. ■

Define

$$\begin{aligned}E(\sigma) &:= C_v \frac{\epsilon^{3/2}}{\sigma^3}, \quad E_0 := E(\sigma_0), \\ T_0 &:= \frac{\epsilon^{3/2}}{\sigma_0}, \quad h_0 := \max\{\|\hat{g}_0\|, \epsilon\}, \\ \mathcal{L}_0 &:= \mathcal{L}_\epsilon(h_0), \quad D_0 := 1 + \log_2 \left(1 + \frac{5C_g}{4\sigma_0^2}\right).\end{aligned}\tag{121}$$

Set

$$B_\epsilon := \Delta_P + \frac{23}{7}E_0 + \frac{T_0}{512}(\mathcal{L}_0 + D_0).\tag{122}$$

For sufficiently small ϵ , let

$$\sigma_* := \sqrt{M_P/6}.$$

Lemma 28 implies that every failed trial occurs below σ_* . Hence, for every finite prefix,

$$J \leq \lceil \log_2(\sigma_* \mathcal{A}_\epsilon) \rceil, \quad \sigma \leq \bar{\sigma} := 2\sigma_*.\tag{123}$$

Lemma 37 (Cumulative accepted-step budget) *Every finite prefix satisfies*

$$\sum_{\text{accepted}} \left(\frac{W_k}{32} + \frac{T_k}{1024} + \frac{U_k}{16} \right) \leq B_\epsilon.\tag{124}$$

If K denotes its number of accepted steps, then

$$K \leq 1024 \bar{\sigma} B_\epsilon \epsilon^{-3/2}, \quad \sum_{\text{accepted}} \sigma_k^2 r_k^3 \leq 8B_\epsilon.\tag{125}$$

Every stored iterate satisfies

$$P(x) - P^* \leq B_\epsilon,$$

and

$$B_\epsilon = \Delta_P + O_{\text{fixed}}(\epsilon^{3/2} \mathcal{A}_\epsilon^3).\tag{126}$$

Proof When the scale is refreshed from σ to 2σ at an unchanged outer point, (78) bounds the corrected-value jump by

$$E(\sigma) + E(2\sigma) = \frac{9}{8}E(\sigma).$$

The failure scales form a doubling sequence, so the sum of all positive corrected-value jumps is at most

$$\frac{9}{8}E_0 \sum_{j \geq 0} 2^{-3j} = \frac{9}{7}E_0.$$

For the gradient-level contribution, the two corrected-gradient errors imply

$$\|\hat{g}_{\text{new}} - \hat{g}_{\text{old}}\| \leq \frac{5C_g \epsilon}{4\sigma^2}.$$

Since $g \mapsto \max\{\|g\|, \epsilon\}$ is 1-Lipschitz,

$$\frac{h_{\text{new}}}{h_{\text{old}}} \leq 1 + \frac{5C_g}{4\sigma^2},$$

and therefore

$$\mathcal{L}_\epsilon(h_{\text{new}}) - \mathcal{L}_\epsilon(h_{\text{old}}) \leq D_0.$$

Because the corresponding $T = \epsilon^{3/2}/\sigma$ is halved after scale doubling, the positive jump of the level term is at most

$$\frac{T_{\text{old}}D_0}{1024}.$$

Summing the doubling sequence gives a total at most

$$\frac{T_0D_0}{512}.$$

The initial potential is bounded above by

$$P(x_0) + E_0 + \frac{T_0}{512}\mathcal{L}_0,$$

while every stored potential is bounded below by $P^* - E_0$. Telescoping Lemma 36 and paying the two types of positive scale-refresh jumps gives

$$\sum_{\text{accepted}} \left(\frac{W}{32} + \frac{T}{1024} + \frac{U}{16} \right) \leq \Delta_P + 2E_0 + \frac{9}{7}E_0 + \frac{T_0}{512}(\mathcal{L}_0 + D_0) = B_\epsilon.$$

This proves (124).

Since

$$T_k \geq \frac{\epsilon^{3/2}}{\sigma},$$

the first estimate in (125) follows. The radius constraint gives

$$4\sigma_k^2 r_k^3 \leq \sigma_k \sqrt{h_k} r_k^2 = W_k,$$

which proves the second estimate.

Finally,

$$E_0 = C_v \epsilon^{3/2} \mathcal{A}_\epsilon^3, \quad T_0 = \epsilon^{3/2} \mathcal{A}_\epsilon.$$

At the initial stored point,

$$\|\hat{g}_0\| \leq \|\nabla P(x_0)\| + C_g \epsilon \mathcal{A}_\epsilon^2.$$

For every fixed payoff and initialization,

$$\mathcal{L}_0 = O_{\text{fixed}}(\mathcal{A}_\epsilon),$$

and

$$D_0 = O_{\text{fixed}}(1 + \log \mathcal{A}_\epsilon).$$

Substitution into (122) gives (126). ■

Proof of Theorem 29 Choose ϵ_0 so that Lemmas 27 and 28 hold and

$$\mathcal{A}_\epsilon^{-1} < \sqrt{M_P/6}$$

for every $0 < \epsilon \leq \epsilon_0$. These restrictions depend only on the common constants (ℓ, μ, ρ) .

Every inner sequence is finite by Lemma 25. Every nonterminal outer trial is either accepted or increases σ . The finite-prefix bounds (123) and (125) therefore imply finite termination.

Let K and J denote the numbers of accepted steps and failed trials. Then

$$Q = K + J + 1,$$

and

$$Q \leq 1024\bar{\sigma}B_\epsilon\epsilon^{-3/2} + \lceil \log_2(\sigma_*\mathcal{A}_\epsilon) \rceil + 1.$$

Using (126) and

$$\bar{\sigma} = 2\sqrt{M_P/6}$$

gives (91a). The output guarantee follows from (85) and the scale cap.

It remains to count successful residual halvings. For an accepted step, set

$$z_k := \frac{\sigma_k r_k}{\sqrt{\epsilon}}.$$

Since $c_0 = 2^{-12}$,

$$N_{\text{tr},k} = \lceil \log_2(1 + 2^{14}z_k) \rceil \leq 16 + z_k^3.$$

Therefore

$$\begin{aligned} \sum_{\text{accepted}} N_{\text{tr},k} &\leq 16K + \epsilon^{-3/2} \sum_{\text{accepted}} \sigma_k^3 r_k^3 \\ &\leq 16K + 8\bar{\sigma}B_\epsilon\epsilon^{-3/2}. \end{aligned} \tag{127}$$

To control failed trials, let

$$L_P := (1 + \kappa)\ell.$$

Smoothness of P and its lower bound give

$$\|\nabla P(x)\|^2 \leq 2L_P(P(x) - P^*).$$

Hence every stored iterate satisfies

$$h \leq \bar{h}_\epsilon := \max \left\{ \epsilon, \sqrt{2L_P B_\epsilon} + C_g \epsilon \mathcal{A}_\epsilon^2 \right\} = O_{\text{fixed}}(1).$$

Each failed trial therefore requires at most

$$N_{\text{fail}} := \left\lceil \log_2 \left(1 + \frac{1}{c_0} \sqrt{\frac{\bar{h}_\epsilon}{\epsilon}} \right) \right\rceil = O_{\text{fixed}}(\mathcal{A}_\epsilon)$$

tracking halvings.

A terminal working update uses at most

$$\lceil \log_2(1 + 1/c_0) \rceil = 13$$

halvings, and each failure uses one additional halving for the scale refresh.

Before validation, the working residual is at most

$$\frac{c_0 \ell \sqrt{\epsilon}}{4\sigma},$$

while the validation target is at least

$$c_0 \mu \epsilon^{3/2}.$$

Thus every validation requires at most

$$N_{\text{val}} := \left\lceil \log_2 \max \left\{ 1, \frac{\kappa \mathcal{A}_\epsilon}{4\epsilon} \right\} \right\rceil = O_{\text{fixed}}(\mathcal{A}_\epsilon)$$

additional successful halvings. Initialization similarly uses

$$N_{\text{init}} := \left\lceil \log_2 \max \left\{ 1, \frac{\|\nabla_y f(x_0, y_{-1})\|}{c_0 a_0 r_\epsilon(\sigma_0)} \right\} \right\rceil = O_{\text{fixed}}(\mathcal{A}_\epsilon)$$

halvings.

Consequently,

$$N_{\text{half}} \leq \sum_{\text{accepted}} N_{\text{tr},k} + J N_{\text{fail}} + J + 13 + (J + 1) N_{\text{val}} + N_{\text{init}}.$$

Since

$$J = O_{\text{fixed}}(1 + \log \mathcal{A}_\epsilon),$$

we obtain

$$N_{\text{half}} = O\left(\Delta_P \sqrt{M_P} \epsilon^{-3/2}\right) + O_{\text{fixed}}(\mathcal{A}_\epsilon^3).$$

All inner objectives $q_x = -f(x, \cdot)$ share the same (μ, ℓ) . Applying Lemma 25 once to the complete sequence, including rejected candidates, scale refreshes, and validators, gives

$$N_y \leq (4 + 2\sqrt{10} C_1) \sqrt{\kappa} (N_{\text{half}} + \lceil \log_4 \kappa \rceil) + 2(J + 2). \quad (128)$$

This proves (91b). Finally,

$$\sqrt{M_P} = \Theta(\kappa^{3/2} \sqrt{\rho}), \quad \sqrt{\kappa M_P} = \Theta(\kappa^2 \sqrt{\rho}),$$

which gives (92). ■